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Trijet production in DIS at small x with massive quarks

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Abstract: Using the formalism of the light-cone wave function together with the Colour Glass Condensate effective theory, we compute the quark–antiquark–gluon Fock-space component of the outgoing state of a transversely polarised virtual photon, keeping a non-zero quark mass m_f throughout, and we deduce the leading-order cross-section for trijet ($q\bar{q}g$) production in electron–proton and electron–nucleus collisions at small Bjorken x. The quark mass introduces helicity-flip pieces at both emission vertices, so the effective vertex of the three-parton wave function involves four transverse tensor structures instead of the single structure of the massless case. In the transverse-coordinate representation required to resum the multiple scattering off the dense target, the wave function is controlled by four radial functions which we obtain in closed form, as one-dimensional Feynman-parameter integrals of modified Bessel functions; the instantaneous pieces remain single Bessel functions, governed by the mass-shifted virtuality scale $E_f^2 = Q^2 + m_f^2/\zeta + m_f^2/\eta$. We present the complete trijet cross-section — direct and interference contributions — in terms of colour dipoles and quadrupoles in the large-N_c limit, give its explicit form in the Gaussian approximation for the target averages, and Fourier-transform the elastic part of the colour structure to transverse momentum space, where the collision enters via convolutions with the dipole unintegrated distribution. In the massless limit our results reproduce the known leading-order trijet cross-section. In the soft-gluon limit the cross-section factorises into the leading-order massive dijet cross-section times the real part of the B-JIMWLK evolution: the slow-gluon logarithm is absorbed into the high-energy evolution with a kernel that is unmodified by the quark mass, which only enters the impact factor. These results provide the real-emission ingredient for next-to-leading order studies of forward heavy-quark dijet and dihadron production at the Electron-Ion Collider.

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1 Introduction

One of the main discoveries of the HERA experimental programme on deeply inelastic electron–proton scattering is the rapid rise of the gluon density inside a hadron as one explores smaller and smaller values of Bjorken x at fixed virtuality Q^2 [1]. While anticipated on the basis of the perturbative evolution in QCD with increasing energy, as encoded in the BFKL equation [2, 3], this rise challenges our understanding of QCD scattering in the vicinity of the unitarity limit. The natural solution to this problem, also emerging from perturbative QCD, is the phenomenon of *gluon saturation* [4, 5], which finds its modern formulation in the effective theory for the *Colour Glass Condensate* (CGC) [6, 7, 8]. In this framework, high-energy scattering off the dense gluonic matter is computed in the eikonal approximation, with the target represented by a strong classical colour field (a "shockwave"), and the projectile partons picking up Wilson lines which resum the multiple scattering to all orders. The non-linear rapidity evolution of the Wilson-line correlators is governed by the Balitsky–JIMWLK hierarchy [9, 10, 11, 12, 13, 14, 15, 16], which at large N_c reduces to the closed Balitsky–Kovchegov (BK) equation [9, 17].

Over the last decade, considerable effort has been devoted to promoting the CGC to next-to-leading order (NLO) accuracy, as required for reliable phenomenology at the future Electron-Ion Collider (EIC) [18, 19]. This programme includes the NLO (and collinearly-improved) versions of the evolution equations [20, 21, 22, 23, 24], as well as NLO "impact factors" for an increasing list of processes: inclusive DIS structure functions [25, 26, 27, 28], inclusive dijets [29, 30] and photoproduced dijets [31], dihadrons [32, 33, 34], photon+dijet [35], and single inclusive hadrons [36, 37]. The real NLO corrections to two-particle production are controlled by the leading-order (LO) cross-section for producing a three-parton ($q\bar{q}g$) final state, which has by now been computed by several groups for *massless* quarks: in momentum space using spinor-helicity techniques [38, 39], and in the mixed (transverse coordinate) representation using light-cone wave functions [34]; see also [40, 41] for the corresponding trijet calculations in proton–nucleus collisions and [42] for three-jet photoproduction.

All of the above refers to massless quarks. Yet there are several compelling reasons to keep a non-zero quark mass. First, charm (and bottom) production is a privileged observable at the EIC: heavy-flavour tagging gives experimentally clean access to the quark channel, and the charm structure function $F_2^{c\bar{c}}$ measured at HERA [43] is an important ingredient in global analyses of the dipole amplitude at small x. Indeed, the NLO dipole factorisation for DIS has recently been extended to massive quarks [44, 45, 46], and the resulting fits [47, 48] show that quark masses are essential for a simultaneous description of the light- and heavy-quark HERA data. Similar conclusions hold for exclusive heavy vector-meson production at NLO [49]. Second, the quark mass acts as a physical infrared regulator: it modifies the aligned-jet (anti-collinear) limits of the photon wave function and thereby the interplay between perturbative and non-perturbative physics in the approach to the photoproduction limit $Q^2 \to 0$. Third, for azimuthal correlations — the observable of choice for gluon saturation studies with dijets and dihadrons [50, 51, 52, 53] — the quark mass is a new scale which competes with the saturation momentum Q_s in the broadening of the back-to-back peak, as emphasised by leading-order studies of forward heavy-quark pairs [54].

For two-parton final states, the massive-quark generalisations are by now standard: the $\gamma^*_{T,L} \to q\bar{q}$ light-cone wave functions with masses have long been known, and the LO massive dijet cross-section in the CGC can be found e.g. in [52]. For the three-parton state the situation is different. The tree-level $\gamma^* \to q\bar{q}g$ amplitudes with massive quarks appear, in momentum space, as intermediate results in the NLO structure-function calculations of refs. [44, 46]; however, to our knowledge, the corresponding outgoing wave function in the transverse coordinate representation — the representation in which the multiple scattering off a dense target is naturally resummed — and the ensuing *trijet cross-section with massive quarks* have not been presented in the literature. The purpose of this paper is to fill this gap.

Specifically, we compute the $q\bar{q}g$ Fock-space component of the outgoing state of a transversely polarised virtual photon with massive quarks, using light-cone perturbation theory (LCPT) [55, 56,

57] in the two-component (spinor-matrix) formulation, and we deduce from it the LO trijet cross-section in DIS at small x. The calculation parallels the massless one of ref. [34], whose conventions and strategy we follow closely, but the quark mass brings in qualitatively new elements:

(i) Both emission vertices (the photon decay $\gamma_T^* \to q\bar{q}$ and the gluon emission $q \to qg$ or $\bar{q} \to \bar{q}g$) acquire a helicity-flip piece proportional to m_f, in addition to the usual helicity-conserving piece. The effective vertex for the three-parton wave function therefore contains *four* transverse tensor structures (flip or non-flip at each vertex), instead of the single structure of the massless case.

(ii) The energy denominators involve two different mass scales: the familiar mass-shifted photon scale $\epsilon_f^2 = \zeta(1-\zeta)Q^2 + m_f^2$ at the photon vertex, and a new scale $\omega_f = \xi m_f/(1-\zeta)$ at the gluon-emission vertex, with ζ and ξ the longitudinal momentum fractions of the quark and the gluon. As a consequence, the Fourier transform to transverse coordinates is no longer a single Bessel function: the four radial functions $\mathcal{G}^{(\alpha\beta)}$ ($\alpha, \beta \in \{0, 1\}$) which control the wave function prior to the scattering are one-dimensional Feynman-parameter integrals of modified Bessel functions, cf. eq. (3.8). They degenerate into elementary expressions in all the interesting limits (massless limit, emission after the shockwave, instantaneous pieces).

(iii) The "instantaneous" contributions remain remarkably simple: they are single Bessel functions $K_1(E_f\mathcal{D})$, with the same effective transverse size $\mathcal{D}$ as in the massless case but with the virtuality Q replaced by the mass-shifted scale $E_f^2 = Q^2 + m_f^2/\zeta + m_f^2/\eta$, cf. eq. (3.17).

Armed with the three-parton wave function, we construct the trijet cross-section as the expectation value of three number-density operators. The colour structure is identical to the massless case — for each of the direct and interference topologies it involves the S-matrices of the $q\bar{q}g$ systems in the direct amplitude (DA) and the complex-conjugate amplitude (CCA), which at large N_c reduce to products of fundamental dipoles and quadrupoles — but the "hard factors" multiplying the various colour structures now contain the full set of mass-dependent helicity sums, which we compute explicitly. For convenience of future numerical studies we also give (a) the cross-section in the Gaussian approximation for the CGC averages, where all quadrupoles collapse onto dipoles, and (b) the explicit Fourier transform of the elastic part of the colour structure to transverse momentum space, where the collision enters only via convolutions with the dipole unintegrated gluon distribution.

Our results have been subjected to several checks. All the spinor-matrix identities, energy denominators, Fourier transforms and helicity sums have been verified by independent symbolic and numerical computations. Most importantly, in the massless limit $m_f \to 0$ our wave function and cross-section reduce exactly — including normalisations — to the results of ref. [34], which in turn agree with the earlier momentum-space calculations [38, 39]; and in the soft-gluon limit we recover the real part of the B-JIMWLK evolution of the LO massive dijet cross-section (section 6).

The paper is organised as follows. In section 2 we collect our conventions, the massive matrix elements of the light-cone Hamiltonian, and the energy denominators. In section 3 we construct the $q\bar{q}g$ component of the outgoing wave function: gluon emission by the antiquark is treated in detail, and the quark-emission case is obtained via symmetry transformations. Section 4 contains the trijet cross-section: direct contributions (antiquark and quark) and the quark–antiquark interference, together with the large-N_c colour structures. In section 5 we Fourier-transform the elastic part of the cross-section to momentum space. Section 6 is devoted to the soft-gluon limit: we show that the cross-section factorises into the LO massive dijet cross-section times the real part of the B-JIMWLK evolution, with an emission kernel unmodified by the quark mass. Section 7 discusses the massless limit and the structure of the mass corrections, and section 8 summarises. Three appendices collect the Fourier transforms (A), the complete helicity sums in momentum space (B), and the Gaussian approximation for the colour correlators (C).

2 Setup, conventions and matrix elements

2.1 Kinematics

We use light-cone coordinates $x^\pm = (x^0 \pm x^3)/\sqrt{2}$ and work in the dipole frame, in which the virtual photon is an energetic right mover with four-momentum $q^\mu = \left(q^+, -\frac{Q^2}{2q^+}, \boldsymbol{q}\right)$, while the hadronic target is an ultrarelativistic left mover. The target is treated in the CGC spirit as a Lorentz-contracted shockwave located at $x^+ = 0$: the scattering of the projectile partons is eikonal and described by Wilson lines,

$$V(\boldsymbol{x}) = \mathrm{P}\exp\left[ig\int dx^+ A_a^-(x^+,\boldsymbol{x})\,t^a\right], \qquad U(\boldsymbol{x}) = \mathrm{P}\exp\left[ig\int dx^+ A_a^-(x^+,\boldsymbol{x})\,T^a\right], \tag{2.1}$$

in the fundamental and the adjoint representation, respectively.

The final state of interest consists of a quark, an antiquark and a gluon, with on-shell momenta k_1, k_2 and k_3. We denote their longitudinal momentum fractions and transverse momenta as follows:

$$\begin{array}{lllll} \text{quark:} & \zeta = k_1^+/q^+, & \boldsymbol{k}_1, & \text{mass } m_f, \\ \text{antiquark:} & \eta = 1-\zeta-\xi, & \boldsymbol{k}_2, & \text{mass } m_f, \\ \text{gluon:} & \xi = k_3^+/q^+, & \boldsymbol{k}_3, & \text{massless}, \end{array} \tag{2.2}$$

where f is the quark flavour. The light-cone energy of an on-shell parton with momentum $(k^+, \boldsymbol{k})$ reads $E_q(k) = (\boldsymbol{k}^2 + m_f^2)/2k^+$ for the quark and the antiquark, and $E_g(k) = \boldsymbol{k}^2/2k^+$ for the gluon.

Throughout the paper we shall use the following notations for the "shifted" transverse momenta, which measure the deviation from collinearity at the two emission vertices (first the photon decay $\gamma^* \to q\bar{q}$, then the gluon emission by the antiquark, cf. section 3),

$$\widetilde{\boldsymbol{k}} \equiv \boldsymbol{k} - \zeta\boldsymbol{q}\,, \qquad \widetilde{\boldsymbol{p}} \equiv \boldsymbol{p} - \frac{\xi}{1-\zeta}\,(\boldsymbol{q}-\boldsymbol{k})\,, \tag{2.3}$$

with $\boldsymbol{k}$ ($\boldsymbol{p}$) the transverse momentum of the quark (gluon), and for the mass scales

$$\epsilon_f^2 \equiv \widetilde{Q}^2 + m_f^2 = \zeta(1-\zeta)Q^2 + m_f^2\,, \qquad \omega_f \equiv \frac{\xi\, m_f}{1-\zeta}\,, \qquad E_f^2 \equiv Q^2 + \frac{m_f^2}{\zeta} + \frac{m_f^2}{\eta}\,, \tag{2.4}$$

where $\widetilde{Q}^2 = \zeta(1-\zeta)Q^2$ is the standard effective virtuality of the $q\bar{q}$ pair. Here ϵ_f is the familiar mass-shifted photon scale which controls the decay $\gamma_T^* \to q\bar{q}$, whereas ω_f is a genuinely new scale introduced by the gluon emission off a massive fermion. Finally, it will prove convenient to introduce

$$a \equiv \frac{\xi\,\eta}{\zeta\,(1-\zeta)^2}\,. \tag{2.5}$$

2.2 Matrix elements of the light-cone Hamiltonian with massive quarks

We work in LCPT in the two-component formalism, where the dynamical degrees of freedom of the fermions are described by two-component spinors χ_λ, with $\lambda = \pm 1/2$ the quark helicity, and the interaction vertices involve the Pauli matrices σ^i ($i = 1, 2$ transverse, σ^3 longitudinal). The three matrix elements of the light-cone QCD Hamiltonian that we need are listed below. They are the massive generalisations of those used in [34] (see also [44, 46] for the massive vertices in the four-component conventions).

(i) Photon decay $\gamma_T^* \to q\bar{q}$. For a transverse photon with polarisation index $\lambda \in \{1, 2\}$,

$$\left\langle q^\alpha_{\lambda_1}(k)\,\bar{q}^\beta_{\lambda_2}(p)\right| H_{\gamma\to q\bar{q}}\,|\gamma_\lambda(q)\rangle = (2\pi)^3\,\delta^{(3)}(q-k-p)\,\frac{e\,e_f\,\delta_{\alpha\beta}}{2\sqrt{2q^+}}$$
$$\times\,\chi^\dagger_{\lambda_1}\left[\frac{2q^\lambda}{q^+} - \sigma^\lambda\,\frac{\boldsymbol{\sigma}\cdot\boldsymbol{p} + m_f\,\sigma^3}{p^+} - \frac{\boldsymbol{\sigma}\cdot\boldsymbol{k} - m_f\,\sigma^3}{k^+}\,\sigma^\lambda\right]\chi_{\lambda_2}, \tag{2.6}$$

with α, β fundamental colour indices and $\delta^{(3)}(q-k-p) \equiv \delta(q^+ - k^+ - p^+)\,\delta^{(2)}(\boldsymbol{q}-\boldsymbol{k}-\boldsymbol{p})$. Using the support of the δ-function to set $k^+ = \zeta q^+$, $p^+ = (1-\zeta)q^+$ and introducing $\widetilde{\boldsymbol{k}} = \boldsymbol{k} - \zeta \boldsymbol{q}$, the spinor bracket reduces to

$$\left[\frac{2q^\lambda}{q^+} - \frac{\boldsymbol{\sigma}\cdot\boldsymbol{k} - m_f\,\sigma^3}{k^+}\,\sigma^\lambda - \sigma^\lambda\,\frac{\boldsymbol{\sigma}\cdot(\boldsymbol{q}-\boldsymbol{k}) + m_f\,\sigma^3}{q^+ - k^+}\right] = \frac{\left[(2\zeta-1)\,\delta^{\lambda j} + i\epsilon^{\lambda j}\sigma^3\right]\widetilde{\boldsymbol{k}}^j + m_f\,\sigma^3\,\sigma^\lambda}{\zeta\,(1-\zeta)\,q^+}\,, \tag{2.7}$$

where we used $\sigma^i\sigma^j = \delta^{ij} + i\epsilon^{ij}\sigma^3$ and $\sigma^\lambda\sigma^3 = -\sigma^3\sigma^\lambda$, with ϵ^{ij} the two-dimensional Levi-Civita symbol ($\epsilon^{12} = +1$). Taking the matrix elements between helicity spinors one finds

$$\chi^\dagger_{\lambda_1}\left[(2\zeta-1)\,\delta^{\lambda j} + i\epsilon^{\lambda j}\sigma^3\right]\chi_{\lambda_2} = \varphi^{\lambda j}_{\lambda_1\lambda_2}(\zeta)\,, \qquad \chi^\dagger_{\lambda_1}\,\sigma^3\sigma^\lambda\,\chi_{\lambda_2} = \mu^\lambda_{\lambda_1\lambda_2}\,, \tag{2.8}$$

with

$$\varphi^{ij}_{\lambda_1\lambda}(\zeta) \equiv \delta_{\lambda_1\lambda}\left[(2\zeta-1)\delta^{ij} + 2i\epsilon^{ij}\lambda_1\right], \qquad \mu^i_{\lambda\lambda'} \equiv \delta_{\lambda',-\lambda}\left[2\lambda\,\delta^{i1} - i\,\delta^{i2}\right], \qquad \sum_i \mu^i_{\lambda\lambda'}\mu^{i*}_{\lambda\lambda'} = 2\,\delta_{\lambda',-\lambda}\,. \tag{2.9}$$

The structure of eq. (2.9) exhibits the key new feature brought in by the quark mass: whereas the standard vertex φ conserves the fermion helicity, the mass term $m_f\,\mu^\lambda_{\lambda_1\lambda_2}$ *flips* it ($\lambda_2 = -\lambda_1$). The matrix element (2.6) can thus be written compactly as

$$\left\langle q^\alpha_{\lambda_1}(k)\,\bar{q}^\beta_{\lambda_2}(p)\right|H_{\gamma\to q\bar{q}}\,|\gamma_\lambda(q)\rangle = (2\pi)^3\,\delta^{(3)}(q-k-p)\,\frac{e\,e_f\,\delta_{\alpha\beta}\left[\varphi^{\lambda j}_{\lambda_1\lambda_2}(\zeta)\,\widetilde{\boldsymbol{k}}^j + m_f\,\mu^\lambda_{\lambda_1\lambda_2}\right]}{2\,\sqrt{2\,q^{+3}}\,\zeta\,(1-\zeta)}\,. \tag{2.10}$$

(ii) Gluon emission by the antiquark. The matrix element for $\bar{q}\to\bar{q}g$ reads

$$\left\langle q^{\alpha'}_{\lambda_1'}(k')\,\bar{q}^{\beta'}_{\lambda_2'}(p')\,g^{a'}_{i'}(q')\right|H_{\bar{q}g}\left|q^\alpha_{\lambda_1}(k)\,\bar{q}^\beta_{\lambda_2}(p)\right\rangle = -\,(2\pi)^6\,\delta^{(3)}(k-k')\,\delta^{(3)}(p-p'-q')\,\delta_{\alpha'\alpha}\,\delta_{\lambda_1\lambda_1'}\,\frac{g\,t^{a'}_{\beta\beta'}}{2\sqrt{2\,q'^+}}$$
$$\times\;\chi^\dagger_{\lambda_2}\left[\frac{2\,q'^{i'}}{q'^+} - \sigma^{i'}\,\frac{\boldsymbol{\sigma}\cdot\boldsymbol{p}' + m_f\,\sigma^3}{p'^+} - \frac{\boldsymbol{\sigma}\cdot\boldsymbol{p} + m_f\,\sigma^3}{p^+}\,\sigma^{i'}\right]\chi_{\lambda_2'}\,, \tag{2.11}$$

where the "backwards" ordering of the helicity and colour labels on the antiquark line reflects the fact that the antiquark formally propagates backwards in time (cf. the discussion in [34]). For the kinematics of interest, $p^+ = (1-\zeta)q^+$, $p'^+ = \eta\,q^+$, $q'^+ = \xi\,q^+$, and with $\widetilde{\boldsymbol{p}} = \boldsymbol{q}' - \frac{\xi}{1-\zeta}\,\boldsymbol{p}$, the terms proportional to $\boldsymbol{p}$ cancel and the spinor bracket reduces to

$$\chi^\dagger_\lambda\left[\frac{2\,q'^{i'}}{q'^+} - \sigma^{i'}\,\frac{\boldsymbol{\sigma}\cdot\boldsymbol{p}' + m_f\,\sigma^3}{p'^+} - \frac{\boldsymbol{\sigma}\cdot\boldsymbol{p} + m_f\,\sigma^3}{p^+}\,\sigma^{i'}\right]\chi_{\lambda_2} = \frac{\tau^{i'n}_{\lambda\lambda_2}(\xi,\eta)\,\widetilde{\boldsymbol{p}}^n + \xi\,\omega_f\,\mu^{i'}_{\lambda\lambda_2}}{\xi\,\eta\,q^+}\,, \tag{2.12}$$

with the helicity-conserving vertex

$$\tau^{mn}_{\lambda\lambda_2}(\xi,\eta) \equiv \delta_{\lambda\lambda_2}\left[(2(1-\zeta)-\xi)\,\delta^{mn} + 2i\,\xi\,\epsilon^{mn}\,\lambda\right], \tag{2.13}$$

identical to the massless case [34], and a helicity-flip piece whose coefficient follows from

$$m_f\,\sigma^3\,\sigma^{i'}\left[\frac{1}{p'^+} - \frac{1}{p^+}\right] = \frac{\xi^2\,m_f}{1-\zeta}\,\frac{\sigma^3\sigma^{i'}}{\xi\,\eta\,q^+} = \xi\,\omega_f\,\frac{\sigma^3\sigma^{i'}}{\xi\,\eta\,q^+}\,. \tag{2.14}$$

Notice that the helicity-flip piece of the gluon-emission vertex comes with the scale $\xi\,\omega_f = \xi^2 m_f/(1-\zeta)$: it is suppressed by two powers of ξ at small ξ, so soft gluon emission remains helicity conserving, as it should.

(iii) Instantaneous vertex. The instantaneous piece of the light-cone fermion propagator generates a direct $\gamma^* \to q\bar{q}g$ vertex. For the time ordering in which the gluon is emitted from the antiquark line,

$$\Big\langle q^\alpha_{\lambda_1}(k)\,\bar{q}^\beta_{\lambda_2}(s)\,g^b_j(p)\Big|\,H_{\gamma\to q\bar{q}g}\,\big|\gamma^i(q)\big\rangle = -(2\pi)^3\,\delta^{(3)}(q-k-p-s)\,\frac{g\,e\,e_f\,\delta_{\lambda_1\lambda_2}\,t^b_{\alpha\beta}}{4\,\sqrt{q^+p^+}}\,\frac{\delta^{ij}+2i\,\epsilon^{ij}\lambda_1}{p^++s^+}\,, \tag{2.15}$$

which is not modified by the quark mass (the mass enters the instantaneous diagrams only via the energy denominators).

2.3 Energy denominators

The LCPT energy denominators follow from the light-cone energy differences at the emission vertices. For the photon decay,

$$E_q + E_{\bar{q}} - E_\gamma = \frac{\widetilde{\boldsymbol{k}}^2+\widetilde{Q}^2+m_f^2}{2\zeta(1-\zeta)q^+} = \frac{\widetilde{\boldsymbol{k}}^2+\epsilon_f^2}{2\,\zeta(1-\zeta)\,q^+}\,, \tag{2.16}$$

while for the subsequent gluon emission from the antiquark,

$$\begin{aligned} E_{q_1}+E_{\bar{q}_1}+E_{g_1}-E_q-E_{\bar{q}} &= \frac{\boldsymbol{p}^2}{2q^+\xi}+\frac{(\boldsymbol{q}-\boldsymbol{k}-\boldsymbol{p})^2+m_f^2}{2q^+\eta}-\frac{(\boldsymbol{q}-\boldsymbol{k})^2+m_f^2}{2q^+(1-\zeta)} \\ &= \frac{\big[(1-\zeta)\boldsymbol{p}-\xi(\boldsymbol{q}-\boldsymbol{k})\big]^2}{2\xi(1-\zeta)\eta\,q^+}+\frac{m_f^2}{2q^+}\left(\frac{1}{\eta}-\frac{1}{1-\zeta}\right) = \frac{(1-\zeta)\left(\widetilde{\boldsymbol{p}}^2+\omega_f^2\right)}{2\,\xi\,\eta\,q^+}\,. \end{aligned} \tag{2.17}$$

The denominator for the full three-parton state then involves the sum of the two energy differences above:

$$E_{q_1}+E_{\bar{q}_1}+E_{g_1}-E_\gamma = \frac{\zeta(1-\zeta)^2\left(\widetilde{\boldsymbol{p}}^2+\omega_f^2\right)+\xi\,\eta\left(\widetilde{\boldsymbol{k}}^2+\epsilon_f^2\right)}{2\,\zeta\,(1-\zeta)\,\xi\,\eta\,q^+}\,. \tag{2.18}$$

Both mass scales of eq. (2.4) appear in eq. (2.18), which is ultimately why the Fourier transform to coordinate space will involve the Feynman-parameter integrals of section 3. For the instantaneous pieces, on the other hand, only the particular combination

$$\zeta(1-\zeta)^2\,\omega_f^2+\xi\,\eta\,\epsilon_f^2 = \zeta\,(1-\zeta)\,\xi\,\eta\,E_f^2\,, \qquad E_f^2 = Q^2+\frac{m_f^2}{\zeta}+\frac{m_f^2}{\eta}\,, \tag{2.19}$$

is relevant, which explains the special role of the mass-shifted virtuality E_f. The identity (2.19) is readily verified using $\eta = 1-\zeta-\xi$.

3 The $q\bar{q}g$ component of the massive photon outgoing state

In this section we construct the three-parton Fock-space component $\big|\gamma_T^i\big\rangle_{q\bar{q}g}$ of the outgoing state generated by the evolution and the scattering of an incoming transversely polarised virtual photon. At leading order this component involves two successive branchings — the photon decay $\gamma_T^* \to q\bar{q}$ followed by gluon emission from either the antiquark or the quark — plus the instantaneous topologies in which both happen at the same light-cone time. As in [34], it is enough to compute in detail the case of emission by the antiquark; the quark case then follows from simple symmetry operations (section 3.5). Graphs in which the photon decays *after* crossing the shockwave need not be computed separately: their effect is accounted for by subtracting the no-scattering limit $\hat{S}\to 1$ from the other contributions.

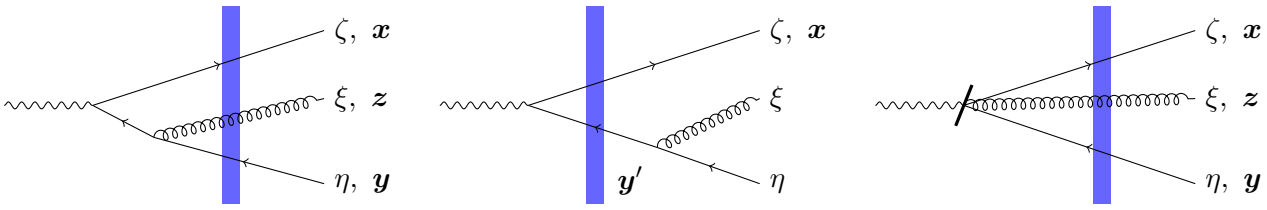


Figure 1: The three topologies contributing to the $q\bar{q}g$ component of the outgoing state for gluon emission by the antiquark: (a) regular emission prior to the shockwave (the blue band), (b) regular emission after the shockwave, (c) the instantaneous vertex, denoted by the short bar. The quark-emission topologies are obtained by exchanging the roles of the quark and the antiquark.

3.1 Gluon emission by the antiquark prior to the shockwave

The contribution of fig. 1.a to the outgoing state is given by second-order LCPT as

$$|\gamma\rangle^{(1)}_{q\bar{q}g} = |q_2\bar{q}_2 g_2\rangle \langle q_2\bar{q}_2 g_2| \hat{S} |q_1\bar{q}_1 g_1\rangle \frac{\langle q_1\bar{q}_1 g_1| H_{\bar{q}g} |q\bar{q}\rangle}{E_{\bar{q}_1} + E_{q_1} + E_{g_1} - E_\gamma} \frac{\langle q\bar{q}| H_{\gamma\to q\bar{q}} |\gamma\rangle}{E_q + E_{\bar{q}} - E_\gamma}\,, \tag{3.1}$$

where sums/integrals over the intermediate states are implicit and $\hat{S}$ is the eikonal S-matrix of the shockwave. We first evaluate eq. (3.1) in the absence of scattering ($\hat{S} = 1$). Using the vertices and energy denominators of section 2, one finds in transverse momentum space

$$\begin{aligned}\big|\gamma_i(Q, q^+, \boldsymbol{q})\big\rangle^{(1a)}_{q\bar{q}g} = -\frac{1}{(2\pi)^6} \int d^2\widetilde{\boldsymbol{k}}\, d^2\widetilde{\boldsymbol{p}} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi \left|q^\alpha_{\lambda_1}(\zeta q^+, \boldsymbol{k})\, \bar{q}^\beta_{\lambda_2}(\eta q^+, \boldsymbol{q}-\boldsymbol{k}-\boldsymbol{p})\, g^a_m(\xi q^+, \boldsymbol{p})\right\rangle \\ \times \frac{e\, e_f\, g\, q^+ \zeta(1-\zeta)}{2\sqrt{\xi}\,\big(\widetilde{\boldsymbol{k}}^2 + \epsilon_f^2\big)} \frac{\Big[\varphi^{ij}_{\lambda_1\lambda}(\zeta)\, \widetilde{\boldsymbol{k}}^j + m_f\, \mu^i_{\lambda_1\lambda}\Big]\Big[\tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\, \widetilde{\boldsymbol{p}}^n + \xi\, \omega_f\, \mu^m_{\lambda\lambda_2}\Big]\, t^a_{\alpha\beta}}{\zeta(1-\zeta)^2\big(\widetilde{\boldsymbol{p}}^2 + \omega_f^2\big) + \xi\eta\big(\widetilde{\boldsymbol{k}}^2 + \epsilon_f^2\big)}\,,\end{aligned} \tag{3.2}$$

with the helicity λ of the intermediate antiquark implicitly summed over. The numerator exhibits the four spin structures announced in the Introduction: non-flip/non-flip ($\varphi\tau$), non-flip/flip ($\varphi\mu$), flip/non-flip ($\mu\tau$) and flip/flip ($\mu\mu$).

Wilson lines are diagonal in the transverse coordinate representation, so we Fourier transform eq. (3.2),

$$\Big|q^\alpha_{\lambda_1}(\boldsymbol{k})\, \bar{q}^\beta_{\lambda_2}(\boldsymbol{q}-\boldsymbol{k}-\boldsymbol{p})\, g^a_m(\boldsymbol{p})\Big\rangle = \int_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}} e^{i\boldsymbol{k}\cdot\boldsymbol{x}}\, e^{i\boldsymbol{p}\cdot\boldsymbol{z}}\, e^{i(\boldsymbol{q}-\boldsymbol{k}-\boldsymbol{p})\cdot\boldsymbol{y}} \Big|q^\alpha_{\lambda_1}(\boldsymbol{x})\, \bar{q}^\beta_{\lambda_2}(\boldsymbol{y})\, g^a_m(\boldsymbol{z})\Big\rangle\,, \tag{3.3}$$

and rewrite the phases in terms of $\widetilde{\boldsymbol{k}}$ and $\widetilde{\boldsymbol{p}}$:

$$e^{i[\zeta\boldsymbol{x}+\eta\boldsymbol{y}+\xi\boldsymbol{z}]\cdot\boldsymbol{q}}\, e^{-i(\boldsymbol{y}-\boldsymbol{z})\cdot\widetilde{\boldsymbol{p}}}\, e^{i\left(\boldsymbol{x}-\frac{\eta\boldsymbol{y}+\xi\boldsymbol{z}}{1-\zeta}\right)\cdot\widetilde{\boldsymbol{k}}} = e^{i\,\boldsymbol{w}\cdot\boldsymbol{q}}\, e^{-i\,\boldsymbol{Y}\cdot\widetilde{\boldsymbol{p}}}\, e^{i\,\boldsymbol{R}\cdot\widetilde{\boldsymbol{k}}}\,. \tag{3.4}$$

The transverse coordinates of the photon ($\boldsymbol{w}$) and of the intermediate antiquark ($\boldsymbol{y}'$, prior to the gluon emission) are fixed by the conservation of the centre of energy at the two vertices,

$$\boldsymbol{w} = \zeta\boldsymbol{x} + (1-\zeta)\,\boldsymbol{y}' = \zeta\boldsymbol{x} + \eta\boldsymbol{y} + \xi\boldsymbol{z}\,, \qquad \boldsymbol{y}' = \frac{\eta\,\boldsymbol{y} + \xi\,\boldsymbol{z}}{1-\zeta}\,, \tag{3.5}$$

and the relevant separations are those between the daughter partons at each vertex,

$$\boldsymbol{R} \equiv \boldsymbol{x} - \boldsymbol{y}'\,, \qquad \boldsymbol{Y} \equiv \boldsymbol{y} - \boldsymbol{z}\,, \qquad (1-\zeta)\,\boldsymbol{x} - \eta\,\boldsymbol{y} - \xi\,\boldsymbol{z} = (1-\zeta)\,\boldsymbol{R}\,. \tag{3.6}$$

In the massless case, the Fourier transform of eq. (3.2) is a single Bessel function [34]. With masses, the two denominators in eq. (3.2) contain two different scales, ϵ_f and ω_f, and there are four numerator structures to transform, one for each helicity path. We therefore define

$$(1-\zeta)\int \frac{d^2\widetilde{\boldsymbol{k}}}{2\pi}\,\frac{d^2\widetilde{\boldsymbol{p}}}{2\pi}\,\frac{e^{i\widetilde{\boldsymbol{k}}\cdot\boldsymbol{R}}\,e^{-i\widetilde{\boldsymbol{p}}\cdot\boldsymbol{Y}}\,\big\{\widetilde{\boldsymbol{k}}^j\widetilde{\boldsymbol{p}}^n,\ \widetilde{\boldsymbol{k}}^j,\ \widetilde{\boldsymbol{p}}^n,\ 1\big\}}{\big(\widetilde{\boldsymbol{k}}^2+\epsilon_f^2\big)\big[\zeta(1-\zeta)^2\big(\widetilde{\boldsymbol{p}}^2+\omega_f^2\big)+\xi\eta\big(\widetilde{\boldsymbol{k}}^2+\epsilon_f^2\big)\big]} = \Big\{R^jY^n\mathcal{G}^{(11)},\ iR^j\mathcal{G}^{(10)},\ -iY^n\mathcal{G}^{(01)},\ \mathcal{G}^{(00)}\Big\}, \tag{3.7}$$

where the functions $\mathcal{G}^{(\alpha\beta)}=\mathcal{G}^{(\alpha\beta)}(R,Y)$ depend only upon the moduli $R=|\boldsymbol{R}|$, $Y=|\boldsymbol{Y}|$ (and on ζ,ξ,Q^2,m_f^2). Combining a Schwinger parametrisation of the two denominators with the Gaussian transverse integrations (the details are given in appendix A), one obtains the four functions as one-dimensional integrals over a Feynman parameter u:

$$\boxed{\mathcal{G}^{(\alpha\beta)}(R,Y)=\frac{a^\beta}{2\,\zeta(1-\zeta)}\int_0^1\frac{du}{u^{1+\beta}}\left(\frac{A_u}{\Delta_u}\right)^{\frac{\alpha+\beta}{2}}K_{\alpha+\beta}\big(\sqrt{A_u\Delta_u}\big)\,,\qquad \alpha,\beta\in\{0,1\}\,,} \tag{3.8}$$

with K_ν the modified Bessel function of the second kind, a as in eq. (2.5), and

$$A_u=\epsilon_f^2+\frac{u\,\omega_f^2}{a}=\zeta(1-\zeta)\,Q^2+m_f^2\left(1+\frac{u\,\zeta\,\xi}{\eta}\right),\qquad \Delta_u=R^2+\frac{a\,Y^2}{u}\,. \tag{3.9}$$

Explicitly,

$$\begin{aligned}
\mathcal{G}^{(11)}&=\frac{a}{2\zeta(1-\zeta)}\int_0^1\frac{du}{u^2}\,\frac{A_u}{\Delta_u}\,K_2\big(\sqrt{A_u\Delta_u}\big)\,,\qquad
\mathcal{G}^{(10)}=\frac{1}{2\zeta(1-\zeta)}\int_0^1\frac{du}{u}\sqrt{\frac{A_u}{\Delta_u}}\,K_1\big(\sqrt{A_u\Delta_u}\big)\,,\\
\mathcal{G}^{(01)}&=\frac{a}{2\zeta(1-\zeta)}\int_0^1\frac{du}{u^2}\sqrt{\frac{A_u}{\Delta_u}}\,K_1\big(\sqrt{A_u\Delta_u}\big)\,,\qquad
\mathcal{G}^{(00)}=\frac{1}{2\zeta(1-\zeta)}\int_0^1\frac{du}{u}\,K_0\big(\sqrt{A_u\Delta_u}\big)\,.
\end{aligned} \tag{3.10}$$

The Feynman parameter u interpolates between the two intrinsic scales of the problem: the endpoint $u\to1$ is controlled by the overall three-parton virtuality, while the endpoint $u\to0$ probes the gluon-emission scale ω_f at large aY^2/u. In the massless limit $\mathcal{G}^{(11)}$ collapses to the single Bessel function of [34] while $\mathcal{G}^{(10)},\mathcal{G}^{(01)},\mathcal{G}^{(00)}$ decouple (their coefficients vanish with m_f); see section 7 and appendix A.

Performing the transform in eq. (3.2) and attaching the Wilson lines appropriate to this topology (all three partons cross the shockwave), and subtracting the no-scattering limit, we arrive at

$$\begin{aligned}
\big|\gamma_i(Q,q^+,\boldsymbol{q})\big\rangle^{(1a)}_{q\bar qg}=-\frac{e\,e_f\,g\,q^+}{2(2\pi)^4}\int_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}}\int_0^1 d\zeta\int_0^{1-\zeta}d\xi\,\Big|q^\alpha_{\lambda_1}(\zeta q^+,\boldsymbol{x})\,\bar q^\beta_{\lambda_2}(\eta q^+,\boldsymbol{y})\,g^a_m(\xi q^+,\boldsymbol{z})\Big\rangle\,\frac{\zeta}{\sqrt{\xi}}\,\delta^{(2)}(\boldsymbol{w}-\boldsymbol{c})\\
\times\,\Psi^{im}_{\lambda_1\lambda_2}[\mathcal{G}](\boldsymbol{R},\boldsymbol{Y})\,\Big[U^{ab}(\boldsymbol{z})\,V(\boldsymbol{x})\,t^b\,V^\dagger(\boldsymbol{y})-t^a\Big]_{\alpha\beta}\,,
\end{aligned} \tag{3.11}$$

where $\boldsymbol{c}=\zeta\boldsymbol{x}+\eta\boldsymbol{y}+\xi\boldsymbol{z}$ is the centre of energy of the trijet system and the "effective vertex" collects the four helicity paths:

$$\begin{aligned}
\Psi^{im}_{\lambda_1\lambda_2}[F]\equiv\sum_\lambda\Big[&\varphi^{ij}_{\lambda_1\lambda}(\zeta)\,\tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\,R^jY^n\,F^{(11)}+i\,\xi\,\omega_f\,\varphi^{ij}_{\lambda_1\lambda}(\zeta)\,\mu^m_{\lambda\lambda_2}\,R^j\,F^{(10)}\\
&-i\,m_f\,\mu^i_{\lambda_1\lambda}\,\tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\,Y^n\,F^{(01)}+m_f\,\xi\,\omega_f\,\mu^i_{\lambda_1\lambda}\,\mu^m_{\lambda\lambda_2}\,F^{(00)}\Big]\,,
\end{aligned} \tag{3.12}$$

for any quadruplet of radial functions $F^{(\alpha\beta)}$.

3.2 Gluon emission after the shockwave

For the topology of fig. 1.b the only differences with respect to the previous subsection are the second energy denominator and the Wilson-line structure. The energy denominator is now the energy difference between the intermediate $q\bar{q}$ pair and the final three-parton state, eq. (2.17), so that

$$\frac{1}{\zeta(1-\zeta)^2\big(\widetilde{\boldsymbol{p}}^2+\omega_f^2\big)+\xi\eta\big(\widetilde{\boldsymbol{k}}^2+\epsilon_f^2\big)} \;\longrightarrow\; -\frac{1}{\zeta(1-\zeta)^2\big(\widetilde{\boldsymbol{p}}^2+\omega_f^2\big)}\,, \tag{3.13}$$

and the two transverse integrations factorise into standard single-scale transforms,

$$\begin{aligned}
\int\frac{d^2\widetilde{\boldsymbol{k}}}{2\pi}\,e^{i\widetilde{\boldsymbol{k}}\cdot\boldsymbol{R}}\,\frac{\{\widetilde{\boldsymbol{k}}^j,1\}}{\widetilde{\boldsymbol{k}}^2+\epsilon_f^2} &= \Big\{\frac{iR^j}{R}\,\epsilon_f\,K_1(\epsilon_f R),\;\; K_0(\epsilon_f R)\Big\},\\
\int\frac{d^2\widetilde{\boldsymbol{p}}}{2\pi}\,e^{-i\widetilde{\boldsymbol{p}}\cdot\boldsymbol{Y}}\,\frac{\{\widetilde{\boldsymbol{p}}^n,1\}}{\widetilde{\boldsymbol{p}}^2+\omega_f^2} &= \Big\{-\frac{iY^n}{Y}\,\omega_f\,K_1(\omega_f Y),\;\; K_0(\omega_f Y)\Big\}.
\end{aligned} \tag{3.14}$$

With the same normalisation as in eq. (3.7), the radial functions for this topology are therefore products of Bessel functions,

$$\begin{aligned}
\mathcal{A}^{(11)} &= \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1-\zeta)R}\,\frac{\omega_f K_1(\omega_f Y)}{Y}\,, \qquad \mathcal{A}^{(10)} = \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1-\zeta)R}\,K_0(\omega_f Y)\,,\\
\mathcal{A}^{(01)} &= \frac{K_0(\epsilon_f R)}{\zeta(1-\zeta)}\,\frac{\omega_f K_1(\omega_f Y)}{Y}\,, \qquad \mathcal{A}^{(00)} = \frac{K_0(\epsilon_f R)}{\zeta(1-\zeta)}\,K_0(\omega_f Y)\,.
\end{aligned} \tag{3.15}$$

Since the gluon is emitted only after the collision, the Wilson lines describe the scattering of the intermediate $q\bar{q}$ pair, with transverse coordinates $\boldsymbol{x}$ and $\boldsymbol{y}'$:

$$\begin{aligned}
|\gamma_i(\boldsymbol{w})\rangle^{(2)}_{q\bar{q}g} = +\frac{e\,e_f\,g\,q^+}{2(2\pi)^4}\int_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}}\int_0^1 d\zeta\int_0^{1-\zeta}d\xi\,\Big|q^\alpha_{\lambda_1}(\zeta q^+,\boldsymbol{x})\,\bar{q}^\beta_{\lambda_2}(\eta q^+,\boldsymbol{y})\,g^a_m(\xi q^+,\boldsymbol{z})\Big\rangle\,\frac{\zeta}{\sqrt{\xi}}\,\delta^{(2)}(\boldsymbol{w}-\boldsymbol{c})\\
\times\,\Psi^{im}_{\lambda_1\lambda_2}[\mathcal{A}](\boldsymbol{R},\boldsymbol{Y})\,\Big[V(\boldsymbol{x})\,V^\dagger(\boldsymbol{y}')\,t^a-t^a\Big]_{\alpha\beta}\,.
\end{aligned} \tag{3.16}$$

Note that the quark mass makes this contribution more singular at large distances than its massless counterpart only in appearance: for $\omega_f Y\ll 1$ one has $\omega_f K_1(\omega_f Y)/Y\to 1/Y^2$, recovering the massless behaviour, whereas for $Y\gtrsim 1/\omega_f$ the emission is exponentially cut off — the scale ω_f acts as the "dead cone" of the massive emitter in transverse coordinate space.

3.3 Instantaneous contribution

For the instantaneous topology of fig. 1.c, the single energy denominator is eq. (2.18) and, by virtue of the identity (2.19), the Fourier transform is again elementary:

$$(1-\zeta)\int\frac{d^2\widetilde{\boldsymbol{k}}}{2\pi}\,\frac{d^2\widetilde{\boldsymbol{p}}}{2\pi}\,\frac{e^{i\widetilde{\boldsymbol{k}}\cdot\boldsymbol{R}}\,e^{-i\widetilde{\boldsymbol{p}}\cdot\boldsymbol{Y}}}{\zeta(1-\zeta)^2\big(\widetilde{\boldsymbol{p}}^2+\omega_f^2\big)+\xi\eta\big(\widetilde{\boldsymbol{k}}^2+\epsilon_f^2\big)} = \frac{E_f\,K_1(E_f\,\mathcal{D})}{\mathcal{D}}\equiv\mathcal{G}_E\,, \tag{3.17}$$

where $\mathcal{D}$ is the same effective transverse size of the three-parton fluctuation as in the massless case,

$$\mathcal{D}^2(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z},\zeta,\xi)=\zeta(1-\zeta)\,\boldsymbol{R}^2+\frac{\xi\eta\,\boldsymbol{Y}^2}{1-\zeta}=\zeta\eta\,(\boldsymbol{x}-\boldsymbol{y})^2+\zeta\xi\,(\boldsymbol{x}-\boldsymbol{z})^2+\eta\xi\,(\boldsymbol{y}-\boldsymbol{z})^2\,, \tag{3.18}$$

but the virtuality which limits this size is now the mass-shifted scale E_f of eq. (2.4): $E_f\,\mathcal{D}\lesssim 1$. The corresponding contribution to the outgoing state reads

$$\begin{aligned}
|\gamma_i(\boldsymbol{w})\rangle^{(1\,\mathrm{inst})}_{q\bar{q}g} = +\frac{e\,e_f\,g\,q^+}{2(2\pi)^4}\int_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}}\int_0^1 d\zeta\int_0^{1-\zeta}d\xi\,\Big|q^\alpha_{\lambda_1}(\zeta q^+,\boldsymbol{x})\,\bar{q}^\beta_{\lambda_2}(\eta q^+,\boldsymbol{y})\,g^a_m(\xi q^+,\boldsymbol{z})\Big\rangle\,\frac{\zeta\,\eta\,\sqrt{\xi}}{1-\zeta}\,\delta^{(2)}(\boldsymbol{w}-\boldsymbol{c})\\
\times\,\big(\delta^{im}+2i\,\epsilon^{im}\lambda_1\big)\,\delta_{\lambda_1\lambda_2}\,\mathcal{G}_E\,\Big[U^{ab}(\boldsymbol{z})\,V(\boldsymbol{x})\,t^b\,V^\dagger(\boldsymbol{y})-t^a\Big]_{\alpha\beta}\,,
\end{aligned} \tag{3.19}$$

with the same Wilson-line structure as in eq. (3.11), as it should (the three partons exist already at the time of the scattering).

3.4 Effective vertices and the combined antiquark-emission wave function

The three contributions above can be packaged into two effective vertices, one for each Wilson-line structure,

$$\begin{aligned}\mathcal{M}^{im}_{1\,\lambda_1\lambda_2}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z},\zeta,\xi) &= \Psi^{im}_{\lambda_1\lambda_2}[\mathcal{G}](\boldsymbol{R},\boldsymbol{Y}) - \delta_{\lambda_1\lambda_2}\,\frac{\xi\eta}{1-\zeta}\left(\delta^{im}+2i\epsilon^{im}\lambda_1\right)\mathcal{G}_E\,,\\ \mathcal{M}^{im}_{2\,\lambda_1\lambda_2}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z},\zeta,\xi) &= \Psi^{im}_{\lambda_1\lambda_2}[\mathcal{A}](\boldsymbol{R},\boldsymbol{Y})\,,\end{aligned} \tag{3.20}$$

in terms of which all antiquark-emission topologies combine into the single expression

$$\begin{aligned}\big|\gamma^i(\boldsymbol{w})\big\rangle_{q\bar{q}g} = -\frac{e\,e_f\,g\,q^+}{2(2\pi)^4}\int_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}}\int_0^1 d\zeta\int_0^{1-\zeta}d\xi\,\Big|q^\alpha_{\lambda_1}(\zeta q^+,\boldsymbol{x})\,\bar{q}^\beta_{\lambda_2}(\eta q^+,\boldsymbol{y})\,g^a_m(\xi q^+,\boldsymbol{z})\Big\rangle\,\frac{\zeta}{\sqrt{\xi}}\,\delta^{(2)}(\boldsymbol{w}-\boldsymbol{c})\\ \times\Big[\mathcal{M}^{im}_{1\,\lambda_1\lambda_2}\Big[U^{ab}(\boldsymbol{z})V(\boldsymbol{x})t^bV^\dagger(\boldsymbol{y})-t^a\Big]_{\alpha\beta}-\mathcal{M}^{im}_{2\,\lambda_1\lambda_2}\Big[V(\boldsymbol{x})\,V^\dagger(\boldsymbol{y}')\,t^a-t^a\Big]_{\alpha\beta}\Big]\,,\end{aligned} \tag{3.21}$$

where the boldface label $\bar{\boldsymbol{q}}$ recalls that the antiquark is the emitter. There is no instantaneous counterpart for the second Wilson structure: the formal substitution $\boldsymbol{y},\boldsymbol{z}\to\boldsymbol{y}'$ which generates the after-shockwave colour structure sends $\boldsymbol{Y}\to 0$, and the instantaneous term vanishes in this limit.

3.5 Gluon emission by the quark

The contribution where the gluon is emitted by the quark need not be computed anew: it follows from eq. (3.21) via symmetry operations which exchange the quark and the antiquark while keeping the final state fixed [34]: (i) exchange $\zeta\leftrightarrow\eta$, $\boldsymbol{x}\leftrightarrow\boldsymbol{y}$, $\alpha\leftrightarrow\beta$, $\lambda_1\leftrightarrow\lambda_2$; (ii) take the complex conjugate of the vertex structures; (iii) flip the overall sign (the colour charge of the quark is minus that of the antiquark). These operations have the following consequences:

- the intermediate coordinate and separations become

$$\boldsymbol{y}'\;\to\;\boldsymbol{x}'=\frac{\zeta\boldsymbol{x}+\xi\boldsymbol{z}}{\zeta+\xi}\,,\qquad \boldsymbol{R}\;\to\;\boldsymbol{y}-\boldsymbol{x}'\equiv-\widetilde{\boldsymbol{R}}\,,\qquad \boldsymbol{Y}\;\to\;\boldsymbol{x}-\boldsymbol{z}\equiv\boldsymbol{X}\,; \tag{3.22}$$

- the spinorial structures transform as

$$\varphi^{ij}_{\lambda_1\lambda}(\zeta)\to\varphi^{ij\,*}_{\lambda_2\lambda}(\eta)=-\varphi^{ij}_{\lambda\lambda_2}(\zeta+\xi)\,,\qquad \tau^{mn}_{\lambda\lambda_2}(\xi,\eta)\to\tau^{mn\,*}_{\lambda\lambda_1}(\xi,\zeta)=\tau^{nm}_{\lambda_1\lambda}(\xi,\zeta)\,, \tag{3.23}$$

  with $\tau^{mn}_{\lambda\lambda'}(\xi,\zeta)=\delta_{\lambda\lambda'}[(2(1-\eta)-\xi)\delta^{mn}+2i\xi\epsilon^{mn}\lambda]$, while the helicity-flip pieces map onto $m_f\,\mu^i_{\lambda\lambda_2}$ and $\xi\,\widetilde{\omega}_f\,\mu^m_{\lambda_1\lambda}$ with

$$\widetilde{\omega}_f=\frac{\xi\,m_f}{1-\eta}\,; \tag{3.24}$$

- the scales and radial functions are obtained by $\zeta\to\eta$ in the explicit factors:

$$\begin{aligned}\widetilde{\epsilon}_f^2&=\eta(1-\eta)Q^2+m_f^2\,,\qquad \widetilde{a}=\frac{\xi\zeta}{\eta(1-\eta)^2}\,,\\ \widetilde{A}_u&=\eta(1-\eta)Q^2+m_f^2\Big(1+\frac{u\,\eta\,\xi}{\zeta}\Big)\,,\qquad \widetilde{\Delta}_u=\widetilde{\boldsymbol{R}}^2+\frac{\widetilde{a}\,\boldsymbol{X}^2}{u}\,,\end{aligned} \tag{3.25}$$

  and $\widetilde{\mathcal{G}}^{(\alpha\beta)}(\widetilde{R},X)$, $\widetilde{\mathcal{A}}^{(\alpha\beta)}(\widetilde{R},X)$ are given by eqs. (3.8) and (3.15) with $(\zeta,a,A_u,\Delta_u,\epsilon_f,\omega_f;R,Y)\to(\eta,\widetilde{a},\widetilde{A}_u,\widetilde{\Delta}_u,\widetilde{\epsilon}_f,\widetilde{\omega}_f;\widetilde{R},X)$. The effective size $\mathcal{D}$ of eq. (3.18) is invariant under these operations: $\mathcal{D}^2=\eta(1-\eta)\widetilde{\boldsymbol{R}}^2+\xi\zeta\boldsymbol{X}^2/(1-\eta)$.

The quark-emission effective vertices read

$$\widetilde{\Psi}^{im}_{\lambda_1\lambda_2}[F] = \sum_{\lambda}\Big[\tau^{nm}_{\lambda_1\lambda}(\xi,\zeta)\,\varphi^{ij}_{\lambda\lambda_2}(\zeta+\xi)\,\widetilde{R}^j\,X^n\,F^{(11)} + i\,\xi\,\widetilde{\omega}_f\,\mu^m_{\lambda_1\lambda}\,\varphi^{ij}_{\lambda\lambda_2}(\zeta+\xi)\,\widetilde{R}^j\,F^{(10)} - i\,m_f\,\tau^{nm}_{\lambda_1\lambda}(\xi,\zeta)\,\mu^i_{\lambda\lambda_2}\,X^n\,F^{(01)} + m_f\,\xi\,\widetilde{\omega}_f\,\mu^m_{\lambda_1\lambda}\,\mu^i_{\lambda\lambda_2}\,F^{(00)}\Big]\,, \tag{3.26}$$

together with

$$\widetilde{\mathcal{M}}^{im}_{1\,\lambda_1\lambda_2} = \widetilde{\Psi}^{im}_{\lambda_1\lambda_2}[\widetilde{\mathcal{G}}](\widetilde{\boldsymbol{R}},\boldsymbol{X}) - \delta_{\lambda_1\lambda_2}\,\frac{\zeta\xi}{\zeta+\xi}\,\big(\delta^{im} - 2i\epsilon^{im}\lambda_1\big)\,\mathcal{G}_E\,, \qquad \widetilde{\mathcal{M}}^{im}_{2\,\lambda_1\lambda_2} = \widetilde{\Psi}^{im}_{\lambda_1\lambda_2}[\widetilde{\mathcal{A}}](\widetilde{\boldsymbol{R}},\boldsymbol{X})\,, \tag{3.27}$$

and the full quark-emission wave function is

$$\big|\gamma^i(\boldsymbol{w})\big\rangle_{\boldsymbol{q}\bar{q}g} = +\frac{e\,e_f\,g\,q^+}{2(2\pi)^4}\int_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}}\int_0^1 d\zeta\int_0^{1-\zeta}d\xi\,\Big|q^\alpha_{\lambda_1}(\zeta q^+,\boldsymbol{x})\,\bar{q}^\beta_{\lambda_2}(\eta q^+,\boldsymbol{y})\,g^a_m(\xi q^+,\boldsymbol{z})\Big\rangle\,\frac{\eta}{\sqrt{\xi}}\,\delta^{(2)}(\boldsymbol{w}-\boldsymbol{c}) \times\Big[\widetilde{\mathcal{M}}^{im}_{1\,\lambda_1\lambda_2}\Big[U^{ab}(\boldsymbol{z})V(\boldsymbol{x})t^bV^\dagger(\boldsymbol{y}) - t^a\Big]_{\alpha\beta} - \widetilde{\mathcal{M}}^{im}_{2\,\lambda_1\lambda_2}\Big[t^a\,V(\boldsymbol{x}')\,V^\dagger(\boldsymbol{y}) - t^a\Big]_{\alpha\beta}\Big]\,. \tag{3.28}$$

Note the position of the colour matrix t^a in the after-shockwave structure: it sits to the left of the Wilson lines when the emitter is the quark, and to the right when it is the antiquark, cf. eq. (3.21). The complete $q\bar{q}g$ component of the outgoing state is the sum of eqs. (3.21) and (3.28).

4 The trijet cross-section at leading order

The leading-order cross-section for $q\bar{q}g$ production is the expectation value of the product of three number-density operators on the outgoing state constructed in the previous section:

$$\frac{d\sigma^{\gamma_T^*A\to q\bar{q}g}}{dk_1^+d^2\boldsymbol{k}_1\,dk_2^+d^2\boldsymbol{k}_2\,dk_3^+d^2\boldsymbol{k}_3}\,(2\pi)\,\delta\big(q^+-k_1^+-k_2^+-k_3^+\big) = \frac{1}{2}\int_{\boldsymbol{w},\bar{\boldsymbol{w}}} e^{-i\boldsymbol{q}\cdot(\boldsymbol{w}-\bar{\boldsymbol{w}})}\;{}_{q\bar{q}g}\big\langle\gamma^i_T(\bar{\boldsymbol{w}})\big|\,\mathcal{N}_q(k_1)\,\mathcal{N}_{\bar{q}}(k_2)\,\mathcal{N}_g(k_3)\,\big|\gamma^i_T(\boldsymbol{w})\big\rangle_{q\bar{q}g}\,, \tag{4.1}$$

where the factor $1/2$ together with the implicit sum over i averages over the transverse polarisations of the photon, and the CGC average over the target field configurations is understood. Below we set $\boldsymbol{q}=0$ for simplicity. As in [34], it is natural to distinguish *direct* contributions, in which the gluon is emitted and reabsorbed by the same fermion, and *interference* contributions, in which it is emitted by the quark in the direct amplitude (DA) and reabsorbed by the antiquark in the complex-conjugate amplitude (CCA), or vice-versa. Transverse coordinates in the CCA carry a bar.

4.1 Direct contribution: emission by the antiquark

A direct calculation using eqs. (4.1) and (3.21) yields

$$\frac{d\sigma^{(\bar{q}),\mathrm{direct}}}{d^3k_1\,d^3k_2\,d^3k_3} = \frac{e^2\,g^2\,\zeta^2\,C_F\,N_c}{8\,(2\pi)^{12}\,\xi\,q^{+2}}\Big(\textstyle\sum_f e_f^2\Big)\,\delta\big(q^+-k_1^+-k_2^+-k_3^+\big)\int_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z},\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}} e^{-i\boldsymbol{k}_1\cdot(\boldsymbol{x}-\bar{\boldsymbol{x}})}\,e^{-i\boldsymbol{k}_2\cdot(\boldsymbol{y}-\bar{\boldsymbol{y}})}\,e^{-i\boldsymbol{k}_3\cdot(\boldsymbol{z}-\bar{\boldsymbol{z}})} \times\Big[\mathcal{H}[\mathcal{G};\bar{\mathcal{G}}]\,\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) - \mathcal{H}[\mathcal{A};\bar{\mathcal{G}}]\,\mathcal{W}(\boldsymbol{x},\boldsymbol{y}',\boldsymbol{y}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) - \mathcal{H}[\mathcal{G};\bar{\mathcal{A}}]\,\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}',\bar{\boldsymbol{y}}') + \mathcal{H}[\mathcal{A};\bar{\mathcal{A}}]\,\mathcal{W}(\boldsymbol{x},\boldsymbol{y}',\boldsymbol{y}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}',\bar{\boldsymbol{y}}')\Big]\,, \tag{4.2}$$

where $d^3k_i \equiv dk_i^+ d^2\boldsymbol{k}_i$, the sum runs over the active quark flavours, and the "hard factors" are the helicity and polarisation sums of products of effective vertices,

$$\mathcal{H}[F;\bar{F}] \equiv \sum_{\lambda_1,\lambda_2} \sum_{l,r} \mathcal{M}^{lr}_{\lambda_1\lambda_2}[F](\boldsymbol{x},\boldsymbol{y},\boldsymbol{z})\, \mathcal{M}^{lr\,*}_{\lambda_1\lambda_2}[\bar{F}](\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})\,, \tag{4.3}$$

with $\mathcal{M}[\mathcal{G}] \equiv \mathcal{M}_1$ and $\mathcal{M}[\mathcal{A}] \equiv \mathcal{M}_2$, cf. eq. (3.20). Performing the sums (we use $\sum_i \mu^i \mu^{i*} = 2\delta_{\lambda',-\lambda}$ and the fact that helicity-conserving and helicity-flip paths do not interfere unless they connect the same pair (λ_1,λ_2)), one finds three types of terms. The regular×regular piece reads

$$\begin{aligned}
\mathcal{H}_{\text{reg}\times\text{reg}}[F;\bar{F}] = R^i Y^m \bar{R}^j \bar{Y}^n \Big\{ & 8\big[\eta^2 + (1-\zeta)^2\big]\big[\zeta^2 + (1-\zeta)^2\big]\,\delta^{ij}\delta^{mn} \\
& - 8\,\xi\,(2\zeta-1)\big(2(1-\zeta)-\xi\big)\,\epsilon^{ij}\epsilon^{mn}\Big\}\, F^{(11)}\bar{F}^{(11)} \\
& + 8\,m_f^2\big[\eta^2 + (1-\zeta)^2\big]\,(\boldsymbol{Y}\cdot\bar{\boldsymbol{Y}})\,F^{(01)}\bar{F}^{(01)} + 8\,\xi^2\omega_f^2\big[\zeta^2 + (1-\zeta)^2\big]\,(\boldsymbol{R}\cdot\bar{\boldsymbol{R}})\,F^{(10)}\bar{F}^{(10)} \\
& + 8\,m_f^2\,\xi^2\omega_f^2\,F^{(00)}\bar{F}^{(00)} \\
& - 8\,\zeta\eta\,m_f\,\xi\,\omega_f\Big[(\boldsymbol{R}\cdot\boldsymbol{Y})\,F^{(11)}\bar{F}^{(00)} + (\bar{\boldsymbol{R}}\cdot\bar{\boldsymbol{Y}})\,F^{(00)}\bar{F}^{(11)} \\
& \qquad + (\boldsymbol{R}\cdot\bar{\boldsymbol{Y}})\,F^{(10)}\bar{F}^{(01)} + (\boldsymbol{Y}\cdot\bar{\boldsymbol{R}})\,F^{(01)}\bar{F}^{(10)}\Big]\,,
\end{aligned} \tag{4.4}$$

the regular×instantaneous interference (plus its hermitian conjugate) is

$$\mathcal{H}_{\text{reg}\times\text{inst}}[F;\bar{F}] = -\frac{8\,\zeta\,\xi\,\eta^2}{1-\zeta}\Big[(\boldsymbol{R}\cdot\boldsymbol{Y})\,F^{(11)}\,\bar{\mathcal{G}}_E + (\bar{\boldsymbol{R}}\cdot\bar{\boldsymbol{Y}})\,\mathcal{G}_E\,\bar{F}^{(11)}\Big] + \frac{8\,\xi^2\,\eta\,m_f\,\omega_f}{1-\zeta}\Big[F^{(00)}\,\bar{\mathcal{G}}_E + \mathcal{G}_E\,\bar{F}^{(00)}\Big]\,, \tag{4.5}$$

and the instantaneous×instantaneous piece is

$$\mathcal{H}_{\text{inst}\times\text{inst}} = 8\left[\frac{\xi\,\eta}{1-\zeta}\right]^2 \mathcal{G}_E\,\bar{\mathcal{G}}_E\,. \tag{4.6}$$

The first line of eq. (4.4) is the massless result of [34] (their eq. (5.5)); all the other terms are new. Note in particular the last line of eq. (4.4), which describes the interference between the double-flip path ($F^{(00)}$) and the non-flip path ($F^{(11)}$), as well as between the two single-flip paths: it is linear in $m_f\,\omega_f \propto m_f^2$, as it must, since a single helicity flip cannot interfere with a non-flip amplitude. Also the second term in eq. (4.5) is a pure mass effect: the instantaneous vertex is helicity conserving, so it now interferes with the double-flip ($F^{(00)}$) path as well.

The effects of the collision are encoded in the function $\mathcal{W}$, defined as the linear combination of partonic S-matrices

$$\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) \equiv S_{q\bar{q}g\,q\bar{q}g}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) - S_{q\bar{q}g}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}) - S^\dagger_{q\bar{q}g}(\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) + 1\,, \tag{4.7}$$

where $S_{q\bar{q}g\,q\bar{q}g}$ describes the scattering of the three-parton system in both the DA and the CCA, and $S_{q\bar{q}g}$ ($S^\dagger_{q\bar{q}g}$) the case where the scattering occurs in the DA (CCA) only. Using Fierz identities to trade the adjoint Wilson lines for fundamental ones, and taking the large-N_c limit, these reduce to products of dipoles and quadrupoles [34],

$$\mathcal{S}(\boldsymbol{x},\boldsymbol{y}) \equiv \frac{1}{N_c}\big\langle \mathrm{tr}\big[V(\boldsymbol{x})V^\dagger(\boldsymbol{y})\big]\big\rangle\,, \qquad \mathcal{Q}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3,\boldsymbol{x}_4) \equiv \frac{1}{N_c}\big\langle \mathrm{tr}\big[V(\boldsymbol{x}_1)V^\dagger(\boldsymbol{x}_2)V(\boldsymbol{x}_3)V^\dagger(\boldsymbol{x}_4)\big]\big\rangle\,, \tag{4.8}$$

namely

$$\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) \simeq \mathcal{Q}(\boldsymbol{x},\boldsymbol{z},\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})\,\mathcal{Q}(\boldsymbol{z},\boldsymbol{y},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) - \mathcal{S}(\boldsymbol{x},\boldsymbol{z})\,\mathcal{S}(\boldsymbol{z},\boldsymbol{y}) - \mathcal{S}(\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})\,\mathcal{S}(\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) + 1\,. \tag{4.9}$$

The "subtracted" colour structures, in which the scattering of the three-parton system in the DA and/or the CCA is replaced by that of the effective $q\bar{q}$ pair, are obtained from eq. (4.9) by the appropriate coincidence limits:

$$\begin{aligned}
\mathcal{W}(\boldsymbol{x},\boldsymbol{y}',\boldsymbol{y}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) &\simeq \mathcal{Q}(\boldsymbol{x},\boldsymbol{y}',\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})\,\mathcal{S}(\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) - \mathcal{S}(\boldsymbol{x},\boldsymbol{y}') - \mathcal{S}(\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})\,\mathcal{S}(\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) + 1\,, \\
\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}',\bar{\boldsymbol{y}}') &\simeq \mathcal{Q}(\boldsymbol{x},\boldsymbol{z},\bar{\boldsymbol{y}}',\bar{\boldsymbol{x}})\,\mathcal{S}(\boldsymbol{z},\boldsymbol{y}) - \mathcal{S}(\boldsymbol{x},\boldsymbol{z})\,\mathcal{S}(\boldsymbol{z},\boldsymbol{y}) - \mathcal{S}(\bar{\boldsymbol{y}}',\bar{\boldsymbol{x}}) + 1\,, \\
\mathcal{W}(\boldsymbol{x},\boldsymbol{y}',\boldsymbol{y}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}',\bar{\boldsymbol{y}}') &\simeq \mathcal{Q}(\boldsymbol{x},\boldsymbol{y}',\bar{\boldsymbol{y}}',\bar{\boldsymbol{x}}) - \mathcal{S}(\boldsymbol{x},\boldsymbol{y}') - \mathcal{S}(\bar{\boldsymbol{y}}',\bar{\boldsymbol{x}}) + 1\,.
\end{aligned} \tag{4.10}$$

The last structure is the same as for the leading-order $q\bar{q}$ production, as expected: it corresponds to graphs where the gluon is emitted and reabsorbed after the collision in both the DA and the CCA. The quark mass does not modify any of these colour structures — it only changes the hard factors which multiply them.

4.2 Direct contribution: emission by the quark

The direct contribution where the gluon is emitted and reabsorbed by the quark follows from the symmetry operations of section 3.5:

$$\begin{aligned}
\frac{d\sigma^{(q),\mathrm{direct}}}{d^3k_1\,d^3k_2\,d^3k_3} &= \frac{e^2\,g^2\,\eta^2\,C_F\,N_c}{8\,(2\pi)^{12}\,\xi\,q^{+2}}\,\Big(\textstyle\sum_f e_f^2\Big)\,\delta\big(q^+-k_1^+-k_2^+-k_3^+\big)\int_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z},\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}} e^{-i\boldsymbol{k}_1\cdot(\boldsymbol{x}-\bar{\boldsymbol{x}})}\,e^{-i\boldsymbol{k}_2\cdot(\boldsymbol{y}-\bar{\boldsymbol{y}})}\,e^{-i\boldsymbol{k}_3\cdot(\boldsymbol{z}-\bar{\boldsymbol{z}})} \\
&\quad\times\Big[\widetilde{\mathcal{H}}[\widetilde{\mathcal{G}};\bar{\widetilde{\mathcal{G}}}]\,\mathcal{W}(\boldsymbol{y},\boldsymbol{x},\boldsymbol{z};\bar{\boldsymbol{y}},\bar{\boldsymbol{x}},\bar{\boldsymbol{z}}) - \widetilde{\mathcal{H}}[\widetilde{\mathcal{A}};\bar{\widetilde{\mathcal{G}}}]\,\mathcal{W}(\boldsymbol{y},\boldsymbol{x}',\boldsymbol{x}';\bar{\boldsymbol{y}},\bar{\boldsymbol{x}},\bar{\boldsymbol{z}}) \\
&\qquad - \widetilde{\mathcal{H}}[\widetilde{\mathcal{G}};\bar{\widetilde{\mathcal{A}}}]\,\mathcal{W}(\boldsymbol{y},\boldsymbol{x},\boldsymbol{z};\bar{\boldsymbol{y}},\bar{\boldsymbol{x}}',\bar{\boldsymbol{x}}') + \widetilde{\mathcal{H}}[\widetilde{\mathcal{A}};\bar{\widetilde{\mathcal{A}}}]\,\mathcal{W}(\boldsymbol{y},\boldsymbol{x}',\boldsymbol{x}';\bar{\boldsymbol{y}},\bar{\boldsymbol{x}}',\bar{\boldsymbol{x}}')\Big],
\end{aligned} \tag{4.11}$$

with $\widetilde{\mathcal{H}}$ built from the vertices (3.27) in full analogy with eq. (4.3). The explicit sums read

$$\begin{aligned}
\widetilde{\mathcal{H}}_{\mathrm{reg}\times\mathrm{reg}} &= \widetilde{R}^i X^m \bar{\widetilde{R}}^j \bar{X}^n \Big\{ 8\big[\zeta^2+(1-\eta)^2\big]\big[\eta^2+(1-\eta)^2\big]\,\delta^{ij}\delta^{mn} \\
&\qquad - 8\,\xi\,(2\eta-1)\big(2(1-\eta)-\xi\big)\,\epsilon^{ij}\epsilon^{mn}\Big\}\,F^{(11)}\bar{F}^{(11)} \\
&\quad + 8m_f^2\big[\zeta^2+(1-\eta)^2\big](\boldsymbol{X}\cdot\bar{\boldsymbol{X}})\,F^{(01)}\bar{F}^{(01)} + 8\xi^2\widetilde{\omega}_f^2\big[\eta^2+(1-\eta)^2\big](\widetilde{\boldsymbol{R}}\cdot\bar{\widetilde{\boldsymbol{R}}})\,F^{(10)}\bar{F}^{(10)} \\
&\quad + 8m_f^2\xi^2\widetilde{\omega}_f^2\,F^{(00)}\bar{F}^{(00)} \\
&\quad + 8\,\zeta\eta\,m_f\,\xi\,\widetilde{\omega}_f\,\Big[(\widetilde{\boldsymbol{R}}\cdot\boldsymbol{X})\,F^{(11)}\bar{F}^{(00)} + (\bar{\widetilde{\boldsymbol{R}}}\cdot\bar{\boldsymbol{X}})\,F^{(00)}\bar{F}^{(11)} \\
&\qquad + (\widetilde{\boldsymbol{R}}\cdot\bar{\boldsymbol{X}})\,F^{(10)}\bar{F}^{(01)} + (\boldsymbol{X}\cdot\bar{\widetilde{\boldsymbol{R}}})\,F^{(01)}\bar{F}^{(10)}\Big]\,,
\end{aligned} \tag{4.12}$$

$$\begin{aligned}
\widetilde{\mathcal{H}}_{\mathrm{reg}\times\mathrm{inst}} &= \frac{8\,\zeta^2\,\xi\,\eta}{1-\eta}\Big[(\widetilde{\boldsymbol{R}}\cdot\boldsymbol{X})\,F^{(11)}\,\bar{\mathcal{G}}_E + (\bar{\widetilde{\boldsymbol{R}}}\cdot\bar{\boldsymbol{X}})\,\mathcal{G}_E\,\bar{F}^{(11)}\Big] + \frac{8\,\xi^2\,\zeta\,m_f\,\widetilde{\omega}_f}{1-\eta}\Big[F^{(00)}\,\bar{\mathcal{G}}_E + \mathcal{G}_E\,\bar{F}^{(00)}\Big]\,, \\
\widetilde{\mathcal{H}}_{\mathrm{inst}\times\mathrm{inst}} &= 8\left[\frac{\xi\,\zeta}{1-\eta}\right]^2\mathcal{G}_E\,\bar{\mathcal{G}}_E\,,
\end{aligned} \tag{4.13}$$

where it is understood that $F^{(\alpha\beta)} = \widetilde{\mathcal{G}}^{(\alpha\beta)}$ or $\widetilde{\mathcal{A}}^{(\alpha\beta)}$ according to the term considered in eq. (4.11). Observe the sign flip of the $F^{(11)}\bar{F}^{(00)}$-type interference terms relative to the antiquark case, eq. (4.4): it originates from the complex conjugation of the helicity structures under the quark–antiquark exchange. The colour structures are obtained from eqs. (4.9)–(4.10) by the relabelling appropriate to a gluon emitted off the quark line, e.g.

$$\mathcal{W}(\boldsymbol{y},\boldsymbol{x},\boldsymbol{z};\bar{\boldsymbol{y}},\bar{\boldsymbol{x}},\bar{\boldsymbol{z}}) \simeq \mathcal{Q}(\boldsymbol{y},\boldsymbol{z},\bar{\boldsymbol{z}},\bar{\boldsymbol{y}})\,\mathcal{Q}(\boldsymbol{z},\boldsymbol{x},\bar{\boldsymbol{x}},\bar{\boldsymbol{z}}) - \mathcal{S}(\boldsymbol{y},\boldsymbol{z})\,\mathcal{S}(\boldsymbol{z},\boldsymbol{x}) - \mathcal{S}(\bar{\boldsymbol{z}},\bar{\boldsymbol{y}})\,\mathcal{S}(\bar{\boldsymbol{x}},\bar{\boldsymbol{z}}) + 1\,, \tag{4.14}$$

and similarly for the subtracted terms, with $\boldsymbol{y}',\bar{\boldsymbol{y}}'$ replaced by $\boldsymbol{x}',\bar{\boldsymbol{x}}'$.

4.3 Interference contributions

Finally, in the interference terms the gluon is emitted by the quark in the DA and reabsorbed by the antiquark in the CCA (plus the complex-conjugate configuration, accounted for by the factor $2\,\mathcal{R}e$):

$$\begin{aligned}\frac{d\sigma^{\rm inter}}{d^3k_1\,d^3k_2\,d^3k_3} &= -\frac{e^2\,g^2\,\eta\,\zeta\,C_F\,N_c}{8\,(2\pi)^{12}\,\xi\,q^{+2}}\left(\textstyle\sum_f e_f^2\right)\delta\big(q^+-k_1^+-k_2^+-k_3^+\big)\\ &\quad\times 2\,\mathcal{R}e\int_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z},\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}} e^{-i\boldsymbol{k}_1\cdot(\boldsymbol{x}-\bar{\boldsymbol{x}})}\,e^{-i\boldsymbol{k}_2\cdot(\boldsymbol{y}-\bar{\boldsymbol{y}})}\,e^{-i\boldsymbol{k}_3\cdot(\boldsymbol{z}-\bar{\boldsymbol{z}})}\\ &\quad\times\Big[\mathcal{H}^{\times}[\widetilde{\mathcal{G}};\bar{\mathcal{G}}]\,\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})\\ &\qquad-\mathcal{H}^{\times}[\widetilde{\mathcal{A}};\bar{\mathcal{G}}]\,\mathcal{W}(\boldsymbol{x}',\boldsymbol{y},\boldsymbol{x}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})\\ &\qquad-\mathcal{H}^{\times}[\widetilde{\mathcal{G}};\bar{\mathcal{A}}]\,\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}',\bar{\boldsymbol{y}}')\\ &\qquad+\mathcal{H}^{\times}[\widetilde{\mathcal{A}};\bar{\mathcal{A}}]\,\mathcal{W}(\boldsymbol{x}',\boldsymbol{y},\boldsymbol{x}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}',\bar{\boldsymbol{y}}')\Big],\end{aligned}\tag{4.15}$$

with the mixed hard factor

$$\mathcal{H}^{\times}[\widetilde{F};\bar{F}]\equiv\sum_{\lambda_1,\lambda_2}\sum_{l,r}\widetilde{\mathcal{M}}^{lr}_{\lambda_1\lambda_2}[\widetilde{F}](\boldsymbol{x},\boldsymbol{y},\boldsymbol{z})\,\mathcal{M}^{lr\,*}_{\lambda_1\lambda_2}[\bar{F}](\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})\,.\tag{4.16}$$

The helicity sums yield

$$\begin{aligned}\mathcal{H}^{\times}_{\rm reg\times reg} &= \widetilde{R}^i X^m \bar{R}^j \bar{Y}^n\Big\{8\big[1-\xi-2\zeta\eta\big]\big[\xi(1-\xi)+2\zeta\eta\big]\,\delta^{ij}\delta^{mn}\\ &\qquad-8\,\xi\,\big[1-2\zeta-\xi\big]^2\,\epsilon^{ij}\epsilon^{mn}\Big\}\,\widetilde{F}^{(11)}\bar{F}^{(11)}\\ &\quad+8\,m_f^2\,(\xi+2\zeta\eta)\,(\boldsymbol{X}\cdot\bar{\boldsymbol{Y}})\,\widetilde{F}^{(01)}\bar{F}^{(01)}-8\,(\xi+2\eta\zeta)\,\xi^2\,\widetilde{\omega}_f\,\omega_f\,(\widetilde{\boldsymbol{R}}\cdot\bar{\boldsymbol{R}})\,\widetilde{F}^{(10)}\bar{F}^{(10)}\\ &\quad-8\,(\zeta+\xi)^2\,m_f\,\xi\,\omega_f\,(\widetilde{\boldsymbol{R}}\cdot\boldsymbol{X})\,\widetilde{F}^{(11)}\bar{F}^{(00)}+8\,(1-\zeta)^2\,m_f\,\xi\,\widetilde{\omega}_f\,(\bar{\boldsymbol{R}}\cdot\bar{\boldsymbol{Y}})\,\widetilde{F}^{(00)}\bar{F}^{(11)}\\ &\quad+8\,\eta^2\,m_f\,\xi\,\widetilde{\omega}_f\,(\widetilde{\boldsymbol{R}}\cdot\bar{\boldsymbol{Y}})\,\widetilde{F}^{(10)}\,\bar{F}^{(01)}-8\,\zeta^2\,m_f\,\xi\,\omega_f\,(\boldsymbol{X}\cdot\bar{\boldsymbol{R}})\,\widetilde{F}^{(01)}\bar{F}^{(10)}\,,\end{aligned}\tag{4.17}$$

$$\mathcal{H}^{\times}_{\rm reg\times inst}=-\frac{8\,\xi\,\eta\,(\xi+\zeta)^2}{1-\zeta}\,(\widetilde{\boldsymbol{R}}\cdot\boldsymbol{X})\,\widetilde{F}^{(11)}\,\bar{\mathcal{G}}_E+\frac{8\,\xi\,\zeta\,(1-\zeta)^2}{\xi+\zeta}\,(\bar{\boldsymbol{R}}\cdot\bar{\boldsymbol{Y}})\,\mathcal{G}_E\,\bar{F}^{(11)}\,,\qquad\mathcal{H}^{\times}_{\rm inst\times inst}=0\,,\tag{4.18}$$

the vanishing of the instantaneous×instantaneous interference being a consequence of the opposite transverse-spin structures, $\sum_{l,r}(\delta^{lr}-2i\epsilon^{lr}\lambda_1)(\delta^{lr}+2i\epsilon^{lr}\lambda_1)^*=0$ for $\lambda_1=\pm1/2$. The massless piece (first line of eq. (4.17)) coincides with eq. (5.17) of [34]. Note that in the interference the mass terms are no longer organised in pairs: e.g. the $\widetilde{F}^{(11)}\bar{F}^{(00)}$ coefficient involves ω_f (the antiquark-side scale) while $\widetilde{F}^{(00)}\bar{F}^{(11)}$ involves $\widetilde{\omega}_f$, reflecting the different emitters in the DA and the CCA.

The colour structures in eq. (4.15) are again products of dipoles and quadrupoles at large N_c:

$$\begin{aligned}\mathcal{W}(\boldsymbol{x}',\boldsymbol{y},\boldsymbol{x}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) &\simeq \mathcal{S}(\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})\,\mathcal{Q}(\boldsymbol{x}',\boldsymbol{y},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})-\mathcal{S}(\boldsymbol{x}',\boldsymbol{y})-\mathcal{S}(\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})\,\mathcal{S}(\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})+1\,,\\ \mathcal{W}(\boldsymbol{x}',\boldsymbol{y},\boldsymbol{x}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}',\bar{\boldsymbol{y}}') &\simeq \mathcal{S}(\bar{\boldsymbol{y}}',\bar{\boldsymbol{x}})\,\mathcal{S}(\boldsymbol{x}',\boldsymbol{y})-\mathcal{S}(\boldsymbol{x}',\boldsymbol{y})-\mathcal{S}(\bar{\boldsymbol{y}}',\bar{\boldsymbol{x}})+1\,,\end{aligned}\tag{4.19}$$

together with $\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})$ and $\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}',\bar{\boldsymbol{y}}')$ already shown in eqs. (4.9)–(4.10). For the numerical evaluation of all these colour correlators one may rely on the Gaussian approximation for the CGC averages, in which each quadrupole is expressed in terms of dipoles only; the complete set of the resulting expressions is collected in appendix C.

5 The cross-section in momentum space: elastic contribution

For many purposes — in particular for studying azimuthal correlations and the approach to the back-to-back limit — it is useful to have the cross-section directly in transverse momentum space. In this section we perform the Fourier transform explicitly for the *elastic* part of the colour structure, i.e. the piece of eqs. (4.9), (4.10) and (4.19) in which each quadrupole is replaced by the product of the dipole built from its DA coordinates and the dipole built from its CCA coordinates, $\mathcal{Q}(\boldsymbol{x},\boldsymbol{z},\bar{\boldsymbol{z}},\bar{\boldsymbol{x}}) \to \mathcal{S}(\boldsymbol{x},\boldsymbol{z})\,\mathcal{S}(\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})$, etc. This is the leading term of the Gaussian decomposition of appendix C, for which the DA and the CCA scatter independently and the six-point colour structure factorises,

$$\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})\Big|_{\rm el} = \big[\mathcal{S}(\boldsymbol{x},\boldsymbol{z})\,\mathcal{S}(\boldsymbol{z},\boldsymbol{y})-1\big]\big[\mathcal{S}(\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})\,\mathcal{S}(\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})-1\big], \tag{5.1}$$

and similarly for the other thirteen colour structures (the subtracted terms factorise exactly in the same way, with the appropriate coincidence limits). For a homogeneous target of transverse area $S_\perp$, the dipole depends only on the coordinate difference and we define

$$\mathcal{S}(\boldsymbol{a},\boldsymbol{b}) = \int \frac{d^2\boldsymbol{l}}{(2\pi)^2}\, e^{i\boldsymbol{l}\cdot(\boldsymbol{a}-\boldsymbol{b})}\,\mathcal{D}(\boldsymbol{l})\,, \qquad \mathcal{D}(\boldsymbol{l}) = \int d^2\boldsymbol{r}\, e^{-i\boldsymbol{l}\cdot\boldsymbol{r}}\,\mathcal{S}(\boldsymbol{r})\,, \qquad \int \frac{d^2\boldsymbol{l}}{(2\pi)^2}\,\mathcal{D}(\boldsymbol{l}) = \mathcal{S}(0) = 1\,. \tag{5.2}$$

5.1 Change of variables and relative momenta

For the antiquark-emission terms we trade $(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z})$ for $(\boldsymbol{c},\boldsymbol{R},\boldsymbol{Y})$ — the centre of energy and the two relative separations of eqs. (3.5)–(3.6) — with unit Jacobian:

$$\boldsymbol{x} = \boldsymbol{c} + (1-\zeta)\boldsymbol{R}, \qquad \boldsymbol{y} = \boldsymbol{c} - \zeta\boldsymbol{R} + \frac{\xi}{1-\zeta}\boldsymbol{Y}, \qquad \boldsymbol{z} = \boldsymbol{c} - \zeta\boldsymbol{R} - \frac{\eta}{1-\zeta}\boldsymbol{Y}, \qquad \boldsymbol{y}' = \boldsymbol{c} - \zeta\boldsymbol{R}\,, \tag{5.3}$$

so that the dipole arguments become

$$\boldsymbol{x}-\boldsymbol{z} = \boldsymbol{R} + \frac{\eta}{1-\zeta}\,\boldsymbol{Y}, \qquad \boldsymbol{z}-\boldsymbol{y} = -\boldsymbol{Y}, \qquad \boldsymbol{x}-\boldsymbol{y}' = \boldsymbol{R}\,. \tag{5.4}$$

For the quark-emission terms we use instead $(\boldsymbol{c},\widetilde{\boldsymbol{R}},\boldsymbol{X})$, cf. eq. (3.22):

$$\boldsymbol{y} = \boldsymbol{c} - (1-\eta)\widetilde{\boldsymbol{R}}, \qquad \boldsymbol{x} = \boldsymbol{c} + \eta\widetilde{\boldsymbol{R}} + \frac{\xi}{1-\eta}\boldsymbol{X}, \qquad \boldsymbol{z} = \boldsymbol{c} + \eta\widetilde{\boldsymbol{R}} - \frac{\zeta}{1-\eta}\boldsymbol{X}, \qquad \boldsymbol{x}' = \boldsymbol{c} + \eta\widetilde{\boldsymbol{R}}\,, \tag{5.5}$$

with $\boldsymbol{x}-\boldsymbol{z} = \boldsymbol{X}$, $\boldsymbol{z}-\boldsymbol{y} = \widetilde{\boldsymbol{R}} - \frac{\zeta}{1-\eta}\boldsymbol{X}$, $\boldsymbol{x}'-\boldsymbol{y} = \widetilde{\boldsymbol{R}}$. The phase of the cross-section separates accordingly,

$$\boldsymbol{k}_1\cdot\boldsymbol{x} + \boldsymbol{k}_2\cdot\boldsymbol{y} + \boldsymbol{k}_3\cdot\boldsymbol{z} = (\boldsymbol{k}_1+\boldsymbol{k}_2+\boldsymbol{k}_3)\cdot\boldsymbol{c} + \widetilde{\boldsymbol{k}}\cdot\boldsymbol{R} - \widetilde{\boldsymbol{p}}\cdot\boldsymbol{Y} = (\boldsymbol{k}_1+\boldsymbol{k}_2+\boldsymbol{k}_3)\cdot\boldsymbol{c} + \widetilde{\boldsymbol{k}}_q\cdot\widetilde{\boldsymbol{R}} - \widetilde{\boldsymbol{p}}_q\cdot\boldsymbol{X}\,, \tag{5.6}$$

with the relative momenta conjugate to the separations,

$$\begin{aligned} \widetilde{\boldsymbol{k}} &= (1-\zeta)\boldsymbol{k}_1 - \zeta(\boldsymbol{k}_2+\boldsymbol{k}_3)\,, & \widetilde{\boldsymbol{p}} &= \frac{\eta\,\boldsymbol{k}_3 - \xi\,\boldsymbol{k}_2}{1-\zeta}\,, \\ \widetilde{\boldsymbol{k}}_q &= \eta(\boldsymbol{k}_1+\boldsymbol{k}_3) - (1-\eta)\boldsymbol{k}_2\,, & \widetilde{\boldsymbol{p}}_q &= \frac{\zeta\,\boldsymbol{k}_3 - \xi\,\boldsymbol{k}_1}{1-\eta}\,. \end{aligned} \tag{5.7}$$

The integral over $\boldsymbol{c}$ produces the overall transverse momentum conservation (recall that we set $\boldsymbol{q}=0$), while the homogeneity of the target produces the factor $S_\perp$.

5.2 Wave functions in momentum space

Inverting the Fourier transforms which defined the radial functions in section 3, the effective vertices can be written as

$$\mathcal{M}^{lr}_{1,2}(\boldsymbol{R},\boldsymbol{Y}) = \int \frac{d^2\widetilde{\boldsymbol{k}}}{2\pi}\frac{d^2\widetilde{\boldsymbol{p}}}{2\pi}\, e^{i\widetilde{\boldsymbol{k}}\cdot\boldsymbol{R}-i\widetilde{\boldsymbol{p}}\cdot\boldsymbol{Y}}\,\Phi^{lr}_{1,2}(\widetilde{\boldsymbol{k}},\widetilde{\boldsymbol{p}})\,, \tag{5.8}$$

with (helicity labels suppressed)

$$\Phi^{lr}_1(\boldsymbol{k},\boldsymbol{p}) = N^{lr}(\boldsymbol{k},\boldsymbol{p})\,f_1(\boldsymbol{k},\boldsymbol{p}) - \xi\eta\left(\delta^{lr}+2i\epsilon^{lr}\lambda_1\right)\delta_{\lambda_1\lambda_2}\,g(\boldsymbol{k},\boldsymbol{p})\,,\qquad \Phi^{lr}_2(\boldsymbol{k},\boldsymbol{p}) = N^{lr}(\boldsymbol{k},\boldsymbol{p})\,f_2(\boldsymbol{k},\boldsymbol{p})\,, \tag{5.9}$$

where the numerator collects the four spin structures,

$$N^{lr}_{\lambda_1\lambda_2}(\boldsymbol{k},\boldsymbol{p}) = \sum_{\lambda}\left[\varphi^{lj}_{\lambda_1\lambda}(\zeta)\,k^j + m_f\,\mu^l_{\lambda_1\lambda}\right]\left[\tau^{rn}_{\lambda\lambda_2}(\xi,\eta)\,p^n + \xi\,\omega_f\,\mu^r_{\lambda\lambda_2}\right], \tag{5.10}$$

and the scalar propagator functions are

$$\begin{aligned} f_1(\boldsymbol{k},\boldsymbol{p}) &= \frac{1-\zeta}{\left(\boldsymbol{k}^2+\epsilon_f^2\right)\left[\zeta(1-\zeta)^2\left(\boldsymbol{p}^2+\omega_f^2\right)+\xi\eta\left(\boldsymbol{k}^2+\epsilon_f^2\right)\right]}\,,\\ g(\boldsymbol{k},\boldsymbol{p}) &= \frac{1}{\zeta(1-\zeta)^2\left(\boldsymbol{p}^2+\omega_f^2\right)+\xi\eta\left(\boldsymbol{k}^2+\epsilon_f^2\right)}\,,\qquad f_2(\boldsymbol{k},\boldsymbol{p}) = \frac{1}{\zeta(1-\zeta)\left(\boldsymbol{k}^2+\epsilon_f^2\right)\left(\boldsymbol{p}^2+\omega_f^2\right)}\,. \end{aligned} \tag{5.11}$$

The quark-emission vertices $\widetilde{\Phi}^{lr}_{1,2}(\widetilde{\boldsymbol{k}}_q,\widetilde{\boldsymbol{p}}_q)$ have the same structure, with the numerator $\widetilde{N}^{lr}_{\lambda_1\lambda_2} = \sum_\lambda[\tau^{nr}_{\lambda_1\lambda}(\xi,\zeta)p^n + \xi\widetilde{\omega}_f\mu^r_{\lambda_1\lambda}][\varphi^{lj}_{\lambda\lambda_2}(\zeta+\xi)k^j + m_f\mu^l_{\lambda\lambda_2}]$, the instantaneous structure $(\delta^{lr}-2i\epsilon^{lr}\lambda_1)$ with coefficient $\xi\zeta\,\widetilde{g}$, and $\widetilde{f}_1,\widetilde{f}_2,\widetilde{g}$ obtained from eq. (5.11) by $(1-\zeta)\to(1-\eta)$, $\xi\eta\to\xi\zeta$, $\zeta(1-\zeta)\to\eta(1-\eta)$, $\epsilon_f\to\widetilde{\epsilon}_f$, $\omega_f\to\widetilde{\omega}_f$.

5.3 Scattering as convolutions and the reduced amplitudes

In the elastic approximation each dipole factor $\mathcal{S}$ translates, via eq. (5.2), into a convolution of the momentum-space vertex with the distribution $\mathcal{D}(\boldsymbol{l})$. Explicitly,

$$\begin{aligned} \int d^2\boldsymbol{R}\,d^2\boldsymbol{Y}\,e^{-i\widetilde{\boldsymbol{k}}\cdot\boldsymbol{R}+i\widetilde{\boldsymbol{p}}\cdot\boldsymbol{Y}}\,\mathcal{M}_1\,\mathcal{S}(\boldsymbol{x},\boldsymbol{z})\,\mathcal{S}(\boldsymbol{z},\boldsymbol{y}) &= (2\pi)^2\int_{\boldsymbol{l}_1,\boldsymbol{l}_2}\mathcal{D}(\boldsymbol{l}_1)\,\mathcal{D}(\boldsymbol{l}_2)\,\Phi_1\Big(\widetilde{\boldsymbol{k}}-\boldsymbol{l}_1,\,\widetilde{\boldsymbol{p}}+\tfrac{\eta}{1-\zeta}\boldsymbol{l}_1-\boldsymbol{l}_2\Big)\,,\\ \int d^2\boldsymbol{R}\,d^2\boldsymbol{Y}\,e^{-i\widetilde{\boldsymbol{k}}\cdot\boldsymbol{R}+i\widetilde{\boldsymbol{p}}\cdot\boldsymbol{Y}}\,\mathcal{M}_2\,\mathcal{S}(\boldsymbol{x},\boldsymbol{y}') &= (2\pi)^2\int_{\boldsymbol{l}_1}\mathcal{D}(\boldsymbol{l}_1)\,\Phi_2\big(\widetilde{\boldsymbol{k}}-\boldsymbol{l}_1,\,\widetilde{\boldsymbol{p}}\big)\,,\\ \int d^2\widetilde{\boldsymbol{R}}\,d^2\boldsymbol{X}\,e^{-i\widetilde{\boldsymbol{k}}_q\cdot\widetilde{\boldsymbol{R}}+i\widetilde{\boldsymbol{p}}_q\cdot\boldsymbol{X}}\,\widetilde{\mathcal{M}}_1\,\mathcal{S}(\boldsymbol{x},\boldsymbol{z})\,\mathcal{S}(\boldsymbol{z},\boldsymbol{y}) &= (2\pi)^2\int_{\boldsymbol{l}_1,\boldsymbol{l}_2}\mathcal{D}(\boldsymbol{l}_1)\,\mathcal{D}(\boldsymbol{l}_2)\,\widetilde{\Phi}_1\Big(\widetilde{\boldsymbol{k}}_q-\boldsymbol{l}_2,\,\widetilde{\boldsymbol{p}}_q+\boldsymbol{l}_1-\tfrac{\zeta}{1-\eta}\boldsymbol{l}_2\Big)\,,\\ \int d^2\widetilde{\boldsymbol{R}}\,d^2\boldsymbol{X}\,e^{-i\widetilde{\boldsymbol{k}}_q\cdot\widetilde{\boldsymbol{R}}+i\widetilde{\boldsymbol{p}}_q\cdot\boldsymbol{X}}\,\widetilde{\mathcal{M}}_2\,\mathcal{S}(\boldsymbol{x}',\boldsymbol{y}) &= (2\pi)^2\int_{\boldsymbol{l}_2}\mathcal{D}(\boldsymbol{l}_2)\,\widetilde{\Phi}_2\big(\widetilde{\boldsymbol{k}}_q-\boldsymbol{l}_2,\,\widetilde{\boldsymbol{p}}_q\big)\,, \end{aligned} \tag{5.12}$$

with $\int_{\boldsymbol{l}}\equiv\int d^2\boldsymbol{l}/(2\pi)^2$. The shifted arguments have a transparent physical interpretation: on the support of $\delta^{(2)}(\boldsymbol{k}_1+\boldsymbol{k}_2+\boldsymbol{k}_3)$ they are simply the relative momenta constructed from the parton momenta *prior* to the scattering, $\boldsymbol{k}_1-\boldsymbol{l}_1$, $\boldsymbol{k}_2+\boldsymbol{l}_2$ and $\boldsymbol{k}_3+\boldsymbol{l}_1-\boldsymbol{l}_2$:

$$\begin{aligned} \boldsymbol{\kappa} &\equiv \widetilde{\boldsymbol{k}}-\boldsymbol{l}_1 = \boldsymbol{k}_1-\boldsymbol{l}_1\,,\qquad \boldsymbol{\rho}\equiv\widetilde{\boldsymbol{p}}+\tfrac{\eta}{1-\zeta}\,\boldsymbol{l}_1-\boldsymbol{l}_2 = (\boldsymbol{k}_3+\boldsymbol{l}_1-\boldsymbol{l}_2)+\tfrac{\xi}{1-\zeta}(\boldsymbol{k}_1-\boldsymbol{l}_1)\,,\\ \boldsymbol{\kappa}_q &\equiv \widetilde{\boldsymbol{k}}_q-\boldsymbol{l}_2 = -(\boldsymbol{k}_2+\boldsymbol{l}_2)\,,\qquad \boldsymbol{\rho}_q\equiv\widetilde{\boldsymbol{p}}_q+\boldsymbol{l}_1-\tfrac{\zeta}{1-\eta}\,\boldsymbol{l}_2 = (\boldsymbol{k}_3+\boldsymbol{l}_1-\boldsymbol{l}_2)+\tfrac{\xi}{1-\eta}(\boldsymbol{k}_2+\boldsymbol{l}_2)\,. \end{aligned} \tag{5.13}$$

That is, $\boldsymbol{l}_1$ and $\boldsymbol{l}_2$ are the transverse momenta transferred by the shockwave to the two effective dipoles.

It is now convenient to introduce reduced amplitudes which combine, for each spin structure, the two Wilson-line topologies of the DA together with their no-scattering subtractions (we use $\int_{\boldsymbol{l}}\mathcal{D}(\boldsymbol{l})=1$ to write the single convolutions under the double one):

$$
\begin{aligned}
T^{jn} &= \int_{\boldsymbol{l}_1,\boldsymbol{l}_2}\mathcal{D}(\boldsymbol{l}_1)\mathcal{D}(\boldsymbol{l}_2)\Big[\kappa^j\rho^n\,f_1(\boldsymbol{\kappa},\boldsymbol{\rho})-\kappa^j\widetilde{\boldsymbol{p}}^n\,f_2(\boldsymbol{\kappa},\widetilde{\boldsymbol{p}})\Big]-\widetilde{\boldsymbol{k}}^j\widetilde{\boldsymbol{p}}^n\Big[f_1(\widetilde{\boldsymbol{k}},\widetilde{\boldsymbol{p}})-f_2(\widetilde{\boldsymbol{k}},\widetilde{\boldsymbol{p}})\Big]\,,\\
T^j_k &= \int_{\boldsymbol{l}_1,\boldsymbol{l}_2}\mathcal{D}(\boldsymbol{l}_1)\mathcal{D}(\boldsymbol{l}_2)\Big[\kappa^j\,f_1(\boldsymbol{\kappa},\boldsymbol{\rho})-\kappa^j\,f_2(\boldsymbol{\kappa},\widetilde{\boldsymbol{p}})\Big]-\widetilde{\boldsymbol{k}}^j\Big[f_1(\widetilde{\boldsymbol{k}},\widetilde{\boldsymbol{p}})-f_2(\widetilde{\boldsymbol{k}},\widetilde{\boldsymbol{p}})\Big]\,,\\
T^n_p &= \int_{\boldsymbol{l}_1,\boldsymbol{l}_2}\mathcal{D}(\boldsymbol{l}_1)\mathcal{D}(\boldsymbol{l}_2)\Big[\rho^n\,f_1(\boldsymbol{\kappa},\boldsymbol{\rho})-\widetilde{\boldsymbol{p}}^n\,f_2(\boldsymbol{\kappa},\widetilde{\boldsymbol{p}})\Big]-\widetilde{\boldsymbol{p}}^n\Big[f_1(\widetilde{\boldsymbol{k}},\widetilde{\boldsymbol{p}})-f_2(\widetilde{\boldsymbol{k}},\widetilde{\boldsymbol{p}})\Big]\,,\\
T_0 &= \int_{\boldsymbol{l}_1,\boldsymbol{l}_2}\mathcal{D}(\boldsymbol{l}_1)\mathcal{D}(\boldsymbol{l}_2)\Big[f_1(\boldsymbol{\kappa},\boldsymbol{\rho})-f_2(\boldsymbol{\kappa},\widetilde{\boldsymbol{p}})\Big]-\Big[f_1(\widetilde{\boldsymbol{k}},\widetilde{\boldsymbol{p}})-f_2(\widetilde{\boldsymbol{k}},\widetilde{\boldsymbol{p}})\Big]\,,\\
T_I &= \int_{\boldsymbol{l}_1,\boldsymbol{l}_2}\mathcal{D}(\boldsymbol{l}_1)\mathcal{D}(\boldsymbol{l}_2)\,g(\boldsymbol{\kappa},\boldsymbol{\rho})-g(\widetilde{\boldsymbol{k}},\widetilde{\boldsymbol{p}})\,,
\end{aligned}
\tag{5.14}
$$

and similarly $\widetilde{T}^{jn},\widetilde{T}^j_k,\widetilde{T}^n_p,\widetilde{T}_0,\widetilde{T}_I$ for the quark emission, with $(\boldsymbol{\kappa},\boldsymbol{\rho})\to(\boldsymbol{\kappa}_q,\boldsymbol{\rho}_q)$, $(\widetilde{\boldsymbol{k}},\widetilde{\boldsymbol{p}})\to(\widetilde{\boldsymbol{k}}_q,\widetilde{\boldsymbol{p}}_q)$ and $f_1,f_2,g\to\widetilde{f}_1,\widetilde{f}_2,\widetilde{g}$. In terms of these, the full scattering amplitude for each helicity configuration reads

$$
\begin{aligned}
\mathcal{A}^{lr}_{\lambda_1\lambda_2} = \sum_{\lambda}\Big[&\varphi^{lj}_{\lambda_1\lambda}(\zeta)\tau^{rn}_{\lambda\lambda_2}(\xi,\eta)\,T^{jn}+\xi\omega_f\,\varphi^{lj}_{\lambda_1\lambda}(\zeta)\mu^r_{\lambda\lambda_2}\,T^j_k+m_f\,\mu^l_{\lambda_1\lambda}\tau^{rn}_{\lambda\lambda_2}(\xi,\eta)\,T^n_p+m_f\xi\omega_f\,\mu^l_{\lambda_1\lambda}\mu^r_{\lambda\lambda_2}\,T_0\Big]\\
&-\xi\eta\,\big(\delta^{lr}+2i\epsilon^{lr}\lambda_1\big)\delta_{\lambda_1\lambda_2}\,T_I\,,
\end{aligned}
\tag{5.15}
$$

and analogously for $\widetilde{\mathcal{A}}^{lr}_{\lambda_1\lambda_2}$ with the quark-side structures. The coordinate integrations then reduce to

$$
\begin{aligned}
&\int_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z},\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}} e^{-i\boldsymbol{k}_1\cdot(\boldsymbol{x}-\bar{\boldsymbol{x}})-i\boldsymbol{k}_2\cdot(\boldsymbol{y}-\bar{\boldsymbol{y}})-i\boldsymbol{k}_3\cdot(\boldsymbol{z}-\bar{\boldsymbol{z}})}\,\big[\text{elastic part}\big]\\
&\qquad = S_\perp\,(2\pi)^2\delta^{(2)}(\boldsymbol{k}_1+\boldsymbol{k}_2+\boldsymbol{k}_3)\,(2\pi)^4\sum_{\lambda_1\lambda_2}\sum_{l,r}\mathcal{A}^{lr}_{\lambda_1\lambda_2}\,\mathcal{A}^{lr\,*}_{\lambda_1\lambda_2}\,,
\end{aligned}
\tag{5.16}
$$

for the direct antiquark term, and similarly for the other contributions.

5.4 Final results in momentum space

It remains to perform the helicity and polarisation sums, for which we use the shorthand notations $|T|^2\equiv\sum_{j,n}T^{jn}T^{jn*}$, $\mathrm{tr}\,T\equiv T^{11}+T^{22}$, $T_k\cdot T^*_p\equiv\sum_j T^j_kT^{j*}_p$ and $\epsilon^{ij}\epsilon^{mn}T^{im}U^{jn*}=T^{11}U^{22*}+T^{22}U^{11*}-T^{12}U^{21*}-T^{21}U^{12*}$. For the direct antiquark contribution, $\sum\mathcal{A}\mathcal{A}^*=H^{\text{massless}}+H^{\text{massive}}$ with

$$
\begin{aligned}
H^{\text{massless}} &= 8\big[\zeta^2+(1-\zeta)^2\big]\big[\eta^2+(1-\zeta)^2\big]\,|T|^2-8\,\xi\,(2\zeta-1)\big(2(1-\zeta)-\xi\big)\,\epsilon^{ij}\epsilon^{mn}\,T^{im}T^{jn*}\\
&\quad-16\,\zeta\,\xi\,\eta^2\,\mathcal{R}e\big[\mathrm{tr}\,T\;T^*_I\big]+8\,\xi^2\eta^2\,|T_I|^2\,,\\
H^{\text{massive}} &= 8\,m_f^2\big[\eta^2+(1-\zeta)^2\big]\,T_p\cdot T^*_p+8\,\xi^2\omega_f^2\big[\zeta^2+(1-\zeta)^2\big]\,T_k\cdot T^*_k+8\,m_f^2\,\xi^2\omega_f^2\,|T_0|^2\\
&\quad-16\,\zeta\eta\,m_f\,\xi\,\omega_f\,\mathcal{R}e\big[\mathrm{tr}\,T\;T_0^*-T_k\cdot T^*_p\big]+16\,\xi\eta\,m_f\,\xi\,\omega_f\,\mathcal{R}e\big[T_0\,T^*_I\big]\,,
\end{aligned}
\tag{5.17}
$$

so that

$$
\boxed{\left.\frac{d\sigma^{(\bar{q}),\text{direct}}}{d^3k_1\,d^3k_2\,d^3k_3}\right|_{\text{el}}=\frac{e^2g^2\,\zeta^2\,C_F\,N_c\,S_\perp}{8\,(2\pi)^6\,\xi\,q^{+2}}\Big(\sum_f e_f^2\Big)\,\delta\big(q^+-\textstyle\sum_i k_i^+\big)\,\delta^{(2)}\big(\sum_i\boldsymbol{k}_i\big)\,\Big[H^{\text{massless}}+H^{\text{massive}}\Big].}
\tag{5.18}
$$

For the direct quark contribution, $\sum \widetilde{\mathcal{A}}\widetilde{\mathcal{A}}^* = \widetilde{H}^{\text{massless}} + \widetilde{H}^{\text{massive}}$ with

$$\begin{aligned}
\widetilde{H}^{\text{massless}} &= 8\big[\zeta^2+(1-\eta)^2\big]\big[\eta^2+(1-\eta)^2\big]\,|\widetilde{T}|^2 - 8\,\xi\,(2\eta-1)\big(2(1-\eta)-\xi\big)\,\epsilon^{ij}\epsilon^{mn}\,\widetilde{T}^{im}\widetilde{T}^{jn*} \\
&\quad + 16\,\zeta^2\,\xi\,\eta\,\mathcal{R}e\big[\operatorname{tr}\widetilde{T}\,\widetilde{T}_I^*\big] + 8\,\xi^2\zeta^2\,|\widetilde{T}_I|^2\,, \\
\widetilde{H}^{\text{massive}} &= 8\,m_f^2\big[\zeta^2+(1-\eta)^2\big]\,\widetilde{T}_p\cdot\widetilde{T}_p^* + 8\,\xi^2\widetilde{\omega}_f^2\big[\eta^2+(1-\eta)^2\big]\,\widetilde{T}_k\cdot\widetilde{T}_k^* + 8\,m_f^2\,\xi^2\widetilde{\omega}_f^2\,|\widetilde{T}_0|^2 \\
&\quad + 16\,\zeta\eta\,m_f\,\xi\,\widetilde{\omega}_f\,\mathcal{R}e\big[\operatorname{tr}\widetilde{T}\,\widetilde{T}_0^* - \widetilde{T}_k\cdot\widetilde{T}_p^*\big] + 16\,\xi\zeta\,m_f\,\xi\,\widetilde{\omega}_f\,\mathcal{R}e\big[\widetilde{T}_0\,\widetilde{T}_I^*\big]\,,
\end{aligned} \tag{5.19}$$

and

$$\left.\frac{d\sigma^{(q),\text{direct}}}{d^3k_1\,d^3k_2\,d^3k_3}\right|_{\text{el}} = \frac{e^2g^2\,\eta^2\,C_F N_c\,S_\perp}{8\,(2\pi)^6\,\xi\,q^{+2}}\Big(\textstyle\sum_f e_f^2\Big)\,\delta\big(q^+-\sum_i k_i^+\big)\,\delta^{(2)}\big(\textstyle\sum_i \boldsymbol{k}_i\big)\,\Big[\widetilde{H}^{\text{massless}}+\widetilde{H}^{\text{massive}}\Big]. \tag{5.20}$$

Finally, the interference contribution involves $\sum \widetilde{\mathcal{A}}\mathcal{A}^* = H_0^{\text{massless}} + H_0^{\text{massive}}$,

$$\begin{aligned}
H_0^{\text{massless}} &= 8\big[1-\xi-2\zeta\eta\big]\big[\xi(1-\xi)+2\zeta\eta\big]\sum_{j,n}\widetilde{T}^{jn}T^{jn*} - 8\,\xi\,\big[1-2\zeta-\xi\big]^2\,\epsilon^{ij}\epsilon^{mn}\,\widetilde{T}^{im}T^{jn*} \\
&\quad - 8\,\xi\eta\,(\zeta+\xi)^2\operatorname{tr}\widetilde{T}\,T_I^* + 8\,\xi\zeta\,(1-\zeta)^2\,\widetilde{T}_I\,\operatorname{tr}T^*\,, \\
H_0^{\text{massive}} &= 8\,m_f^2\,(\xi+2\zeta\eta)\,\widetilde{T}_p\cdot T_p^* - 8\,(\xi+2\zeta\eta)\,\xi^2\,\widetilde{\omega}_f\,\omega_f\,\widetilde{T}_k\cdot T_k^* \\
&\quad - 8\,(\zeta+\xi)^2\,m_f\,\xi\,\omega_f\operatorname{tr}\widetilde{T}\,T_0^* + 8\,(1-\zeta)^2\,m_f\,\xi\,\widetilde{\omega}_f\,\widetilde{T}_0\,\operatorname{tr}T^* \\
&\quad - 8\,\eta^2\,m_f\,\xi\,\widetilde{\omega}_f\,\widetilde{T}_k\cdot T_p^* + 8\,\zeta^2\,m_f\,\xi\,\omega_f\,\widetilde{T}_p\cdot T_k^*\,,
\end{aligned} \tag{5.21}$$

and reads

$$\left.\frac{d\sigma^{\text{inter}}}{d^3k_1\,d^3k_2\,d^3k_3}\right|_{\text{el}} = -\frac{e^2g^2\,\eta\,\zeta\,C_F N_c\,S_\perp}{8\,(2\pi)^6\,\xi\,q^{+2}}\Big(\textstyle\sum_f e_f^2\Big)\,\delta\big(q^+-\sum_i k_i^+\big)\,\delta^{(2)}\big(\textstyle\sum_i \boldsymbol{k}_i\big)\;2\,\mathcal{R}e\,\Big[H_0^{\text{massless}}+H_0^{\text{massive}}\Big]. \tag{5.22}$$

Eqs. (5.18), (5.20) and (5.22), with the reduced amplitudes (5.14), are our final momentum-space results. They involve at most two-dimensional convolutions with $\mathcal{D}(\boldsymbol{l})$ and are thus well suited for numerical evaluation. The complete set of helicity contractions needed to re-derive, or generalise, these sums (e.g. for the non-elastic colour structures) is collected in appendix B.

6 The soft-gluon limit: recovering the B-JIMWLK evolution

In this section we study the limit in which the gluon emission is soft, $\xi \to 0$ at fixed ζ, and demonstrate that the trijet cross-section of section 4 reproduces the expected part of the B-JIMWLK evolution of the leading-order (massive) dijet cross-section — namely, the "real" terms, in which the soft gluon crosses the cut. This is both an important consistency check and the key ingredient for the NLO programme: upon integrating out the gluon, the longitudinal integration develops a logarithmic divergence $\int d\xi/\xi$ at small ξ, which must be subtracted and resummed into the high-energy evolution of the LO cross-section. The central result of this section is that *the quark mass does not modify this structure at all*: the soft emission kernels and the colour structures are identical to those of the massless case [34], and the whole mass dependence factorises into the hard factor of the LO massive dijet cross-section.

6.1 Simplifications in the soft limit

For $\xi \to 0$ one encounters several simplifications, in close parallel with the massless discussion in [34], but with some noteworthy mass-specific features.

(a) Kinematics. The recoil of the emitter can be neglected: $\boldsymbol{y}' \simeq \boldsymbol{y}$ for an emission by the antiquark and $\boldsymbol{x}' \simeq \boldsymbol{x}$ for an emission by the quark, hence $\boldsymbol{R} \simeq \widetilde{\boldsymbol{R}} \simeq \boldsymbol{x} - \boldsymbol{y}$. In this section we accordingly write $R \equiv |\boldsymbol{x} - \boldsymbol{y}|$ (and $\bar{R} \equiv |\bar{\boldsymbol{x}} - \bar{\boldsymbol{y}}|$ in the CCA). Likewise $\eta \to 1 - \zeta$, so that $\widetilde{\epsilon}_f \simeq \epsilon_f$, and $\mathcal{D}^2 \simeq \zeta(1-\zeta)(\boldsymbol{x}-\boldsymbol{y})^2$. The two mass scales associated with the gluon vertex disappear in this limit, $\omega_f = \xi m_f/(1-\zeta) \to 0$ and $\widetilde{\omega}_f \to 0$: the "dead cone" of the massive emitter, of transverse size $1/\omega_f \propto 1/\xi$, is pushed to infinity, so soft emission is insensitive to the quark mass — as required by the soft theorems.

(b) Radial functions. Since $a = \xi\eta/[\zeta(1-\zeta)^2] \to 0$, the Feynman-parameter integrals (3.8) can be evaluated by the change of variable $v = aY^2/u$, with the result that the two relevant functions collapse to single Bessel functions,

$$\mathcal{G}^{(11)} \simeq \mathcal{A}^{(11)} \simeq \frac{1}{\boldsymbol{Y}^2}\,\frac{\epsilon_f\, K_1(\epsilon_f R)}{\zeta(1-\zeta)\, R}\,, \qquad \mathcal{G}^{(01)} \simeq \mathcal{A}^{(01)} \simeq \frac{1}{\boldsymbol{Y}^2}\,\frac{K_0(\epsilon_f R)}{\zeta(1-\zeta)}\,, \tag{6.1}$$

up to corrections of relative order $\mathcal{O}(a)$. (For instance, for $\mathcal{G}^{(11)}$ one finds $\int_0^1 (du/u^2)(A_u/\Delta_u) K_2(\sqrt{A_u\Delta_u}) = (1/aY^2)\int_{aY^2}^\infty dv\,[\epsilon_f^2/(R^2+v)] K_2(\epsilon_f\sqrt{R^2+v}) + \mathcal{O}(a)$, and the remaining integral is elementary.) The same limits hold for the after-shockwave functions $\mathcal{A}^{(\alpha\beta)}$, by virtue of $\omega_f K_1(\omega_f Y)/Y \to 1/Y^2$ and the identities (3.15): the distinction between emissions before and after the shockwave disappears from the *kernel*, surviving only in the Wilson-line structure. The analogous statements hold for the quark-emission functions, with $\boldsymbol{Y} \to \boldsymbol{X}$.

(c) Vertices. In the effective vertices (3.12) and (3.26), the structures $F^{(10)}$ and $F^{(00)}$ carry explicit factors $\xi\omega_f \propto \xi^2$ (or $\xi\widetilde{\omega}_f$) and drop out: *a soft gluon cannot flip the helicity of a massive quark*. The gluon-emission vertices become spin diagonal and eikonal, $\tau^{mn}_{\lambda\lambda_2}(\xi,\eta) \to 2(1-\zeta)\,\delta_{\lambda\lambda_2}\,\delta^{mn}$ (its ϵ^{mn} piece is of order ξ), while $\varphi^{ij}_{\lambda\lambda_2}(\zeta+\xi) \to \varphi^{ij}_{\lambda\lambda_2}(\zeta)$. Introducing the coordinate-space vertex of the *leading-order massive* photon decay,

$$\psi^i_{\lambda_1\lambda_2}(\zeta;\boldsymbol{R}) \equiv \varphi^{ij}_{\lambda_1\lambda_2}(\zeta)\,\frac{R^j}{R}\,\epsilon_f K_1(\epsilon_f R) \;-\; i\,m_f\,\mu^i_{\lambda_1\lambda_2}\,K_0(\epsilon_f R)\,, \tag{6.2}$$

the effective vertices factorise into the LO vertex times a universal emission kernel,

$$\Psi^{im}_{\lambda_1\lambda_2}[\mathcal{G}] \simeq \Psi^{im}_{\lambda_1\lambda_2}[\mathcal{A}] \simeq \frac{2}{\zeta}\,\frac{Y^m}{\boldsymbol{Y}^2}\,\psi^i_{\lambda_1\lambda_2}(\zeta;\boldsymbol{R})\,, \qquad \widetilde{\Psi}^{im}_{\lambda_1\lambda_2}[\widetilde{\mathcal{G}}] \simeq \widetilde{\Psi}^{im}_{\lambda_1\lambda_2}[\widetilde{\mathcal{A}}] \simeq \frac{2}{1-\zeta}\,\frac{X^m}{\boldsymbol{X}^2}\,\psi^i_{\lambda_1\lambda_2}(\zeta;\boldsymbol{R})\,. \tag{6.3}$$

(d) Instantaneous terms. The instantaneous pieces of the effective vertices (3.20) and (3.27) carry the explicit small factors $\xi\eta/(1-\zeta)$ and $\zeta\xi/(\zeta+\xi)$, both of order ξ, while $\mathcal{G}_E$ stays finite; hence they do not contribute in the soft limit. This also follows from the energy denominators: for the regular graphs the denominator is dominated by the gluon light-cone energy, $E_{q\bar{q}g} - E_{q\bar{q}} \simeq E_{q\bar{q}g} - E_\gamma \simeq \boldsymbol{p}^2/2\xi q^+$, corresponding to the short formation time $\tau_p \sim 2\xi q^+/\boldsymbol{p}^2$ of the soft gluon, whereas the instantaneous graphs involve the product of two large energy differences and scale like $\tau_p^2 \sim \xi^2$.

Inserting eq. (6.3) into eqs. (3.21) and (3.28), the no-scattering subtraction terms cancel between the before- and after-shockwave topologies, and the soft three-parton state takes the transparent eikonal form

$$\begin{aligned}\big|\gamma^i(\boldsymbol{w})\big\rangle^{\rm soft}_{q\bar{q}g} = -\frac{e\,e_f\,g\,q^+}{(2\pi)^4\sqrt{\xi}}\int_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}}\int_0^1 d\zeta\,\Big|q^\alpha_{\lambda_1}(\boldsymbol{x})\,\bar{q}^\beta_{\lambda_2}(\boldsymbol{y})\,g^a_m(\boldsymbol{z})\Big\rangle\,\delta^{(2)}(\boldsymbol{w}-\boldsymbol{c})\,\psi^i_{\lambda_1\lambda_2}(\zeta;\boldsymbol{R})\\ \times\Big\{\frac{Y^m}{\boldsymbol{Y}^2}\Big[U^{ab}(\boldsymbol{z})V(\boldsymbol{x})t^bV^\dagger(\boldsymbol{y}) - V(\boldsymbol{x})V^\dagger(\boldsymbol{y})\,t^a\Big]_{\alpha\beta}\\ -\frac{X^m}{\boldsymbol{X}^2}\Big[U^{ab}(\boldsymbol{z})V(\boldsymbol{x})t^bV^\dagger(\boldsymbol{y}) - t^a\,V(\boldsymbol{x})V^\dagger(\boldsymbol{y})\Big]_{\alpha\beta}\Big\}\,,\end{aligned} \tag{6.4}$$

where the longitudinal momentum fractions $(\zeta, 1-\zeta, \xi)$ of the three partons are left implicit in the Fock state. Here $\boldsymbol{X} = \boldsymbol{x} - \boldsymbol{z}$ and $\boldsymbol{Y} = \boldsymbol{y} - \boldsymbol{z}$: this is the LO massive $q\bar{q}$ wave function, dressed by

the standard Weizsäcker–Williams emission current of a soft gluon from the antiquark ($Y^m/\boldsymbol{Y}^2$) or the quark ($X^m/\boldsymbol{X}^2$), emitted either before the shockwave (the $UVtV^\dagger$ pieces) or after it (the $VV^\dagger t$ and $tVV^\dagger$ pieces).

6.2 Factorisation of the hard factors

Squaring, the helicity sums of section 4 degenerate: the direct antiquark, direct quark and interference hard factors all become proportional to one and the same LO quantity. Explicitly, one finds (we have verified these limits directly from eqs. (4.4), (4.12) and (4.17))

$$\zeta^2\,\mathcal{H} \;\simeq\; 16\,\frac{\boldsymbol{Y}\cdot\bar{\boldsymbol{Y}}}{\boldsymbol{Y}^2\bar{\boldsymbol{Y}}^2}\,\mathcal{H}^T_{\rm LO}\,, \qquad \eta^2\,\widetilde{\mathcal{H}} \;\simeq\; 16\,\frac{\boldsymbol{X}\cdot\bar{\boldsymbol{X}}}{\boldsymbol{X}^2\bar{\boldsymbol{X}}^2}\,\mathcal{H}^T_{\rm LO}\,, \qquad \zeta\eta\,\mathcal{H}^\times \;\simeq\; 16\,\frac{\boldsymbol{X}\cdot\bar{\boldsymbol{Y}}}{\boldsymbol{X}^2\bar{\boldsymbol{Y}}^2}\,\mathcal{H}^T_{\rm LO}\,, \tag{6.5}$$

with $\bar{\boldsymbol{X}}=\bar{\boldsymbol{x}}-\bar{\boldsymbol{z}}$, $\bar{\boldsymbol{Y}}=\bar{\boldsymbol{y}}-\bar{\boldsymbol{z}}$, and where

$$\boxed{\mathcal{H}^T_{\rm LO}(\zeta;R,\bar{R}) \equiv \left[\zeta^2+(1-\zeta)^2\right]\epsilon_f^2\,\frac{\boldsymbol{R}\cdot\bar{\boldsymbol{R}}}{R\,\bar{R}}\,K_1(\epsilon_f R)\,K_1(\epsilon_f\bar{R}) \;+\; m_f^2\,K_0(\epsilon_f R)\,K_0(\epsilon_f\bar{R})} \tag{6.6}$$

is precisely the hard factor of the LO dijet cross-section for a transverse photon with massive quarks — the familiar combination of the helicity-conserving (K_1^2) and helicity-flip (K_0^2) pieces of the massive photon wave function. Eq. (6.5) is the massive generalisation of eq. (6.1) in [34]: the dependence on ζ and ξ becomes the same for all three channels, the photon decay vertex factorises from the gluon emission, and the only memory of the emitter is the transverse separation ($\boldsymbol{X}$ or $\boldsymbol{Y}$) entering the emission kernel. Note in particular that *both* helicity structures of the LO vertex are multiplied by the *same* soft kernel: the high-energy evolution generated below is diagonal in the helicity structure of the massive LO wave function.

6.3 The soft cross-section and B-JIMWLK

Adding the direct and interference contributions of section 4 with the simplifications above (and neglecting the soft k_3^+ in the longitudinal δ-function), the trijet cross-section takes the factorised form

$$\begin{aligned}\left.\frac{d\sigma^{\gamma_T^* A\to q\bar{q}g}}{d^3k_1\,d^3k_2\,d^3k_3}\right|_{\xi\to 0} &= \frac{2\,\alpha_{em}\,N_c}{(2\pi)^6\,q^+}\left(\sum\nolimits_f e_f^2\right)\delta\big(q^+-k_1^+-k_2^+\big)\int_{\boldsymbol{x},\boldsymbol{y},\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}} e^{-i\boldsymbol{k}_1\cdot(\boldsymbol{x}-\bar{\boldsymbol{x}})-i\boldsymbol{k}_2\cdot(\boldsymbol{y}-\bar{\boldsymbol{y}})}\,\mathcal{H}^T_{\rm LO}(\zeta;R,\bar{R})\\ &\quad\times\frac{4}{(2\pi)^4}\,\frac{\alpha_s C_F}{\xi\,q^+}\int_{\boldsymbol{z},\bar{\boldsymbol{z}}} e^{-i\boldsymbol{k}_3\cdot(\boldsymbol{z}-\bar{\boldsymbol{z}})}\,\Delta_{\rm real}\mathcal{Q}(\boldsymbol{x},\boldsymbol{y},\bar{\boldsymbol{y}},\bar{\boldsymbol{x}};\boldsymbol{z},\bar{\boldsymbol{z}})\,,\end{aligned} \tag{6.7}$$

where the first line is the would-be LO massive dijet cross-section stripped of its S-matrix structure, while the second line encodes the soft emission. For $m_f\to 0$, $\mathcal{H}^T_{\rm LO}\to[\zeta^2+(1-\zeta)^2]\bar{Q}^2K_1(\bar{Q}R)K_1(\bar{Q}\bar{R})(\boldsymbol{R}\cdot\bar{\boldsymbol{R}})/(R\bar{R})$ and eq. (6.7) coincides, including the overall normalisation, with eq. (6.3) of [34]. The function $\Delta_{\rm real}\mathcal{Q}$ collects the emission kernels and the surviving colour structures: writing $\mathcal{K}_{\boldsymbol{X}\bar{\boldsymbol{X}}}\equiv(\boldsymbol{X}\cdot\bar{\boldsymbol{X}})/(\boldsymbol{X}^2\bar{\boldsymbol{X}}^2)$ and similarly for the other pairings, we find

$$\begin{aligned}\Delta_{\rm real}\mathcal{Q}(\boldsymbol{x},\boldsymbol{y},\bar{\boldsymbol{y}},\bar{\boldsymbol{x}};\boldsymbol{z},\bar{\boldsymbol{z}}) \equiv{}& \Big[\mathcal{K}_{\boldsymbol{X}\bar{\boldsymbol{X}}}+\mathcal{K}_{\boldsymbol{Y}\bar{\boldsymbol{Y}}}-\mathcal{K}_{\boldsymbol{X}\bar{\boldsymbol{Y}}}-\mathcal{K}_{\boldsymbol{Y}\bar{\boldsymbol{X}}}\Big]\,\mathcal{Q}(\boldsymbol{x},\boldsymbol{z},\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})\,\mathcal{Q}(\boldsymbol{z},\boldsymbol{y},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})\\ &+\Big[\mathcal{K}_{\boldsymbol{X}\bar{\boldsymbol{X}}}+\mathcal{K}_{\boldsymbol{Y}\bar{\boldsymbol{Y}}}\Big]\,\mathcal{Q}(\boldsymbol{x},\boldsymbol{y},\bar{\boldsymbol{y}},\bar{\boldsymbol{x}})-\Big[\mathcal{K}_{\boldsymbol{X}\bar{\boldsymbol{Y}}}+\mathcal{K}_{\boldsymbol{Y}\bar{\boldsymbol{X}}}\Big]\,\mathcal{S}(\boldsymbol{x},\boldsymbol{y})\,\mathcal{S}(\bar{\boldsymbol{y}},\bar{\boldsymbol{x}})\\ &-\Big[\mathcal{K}_{\boldsymbol{X}\bar{\boldsymbol{X}}}-\mathcal{K}_{\boldsymbol{X}\bar{\boldsymbol{Y}}}\Big]\,\mathcal{Q}(\boldsymbol{x},\boldsymbol{y},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})\,\mathcal{S}(\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})-\Big[\mathcal{K}_{\boldsymbol{Y}\bar{\boldsymbol{Y}}}-\mathcal{K}_{\boldsymbol{Y}\bar{\boldsymbol{X}}}\Big]\,\mathcal{Q}(\boldsymbol{x},\boldsymbol{y},\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})\,\mathcal{S}(\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})\\ &-\Big[\mathcal{K}_{\boldsymbol{X}\bar{\boldsymbol{X}}}-\mathcal{K}_{\boldsymbol{Y}\bar{\boldsymbol{X}}}\Big]\,\mathcal{S}(\boldsymbol{x},\boldsymbol{z})\,\mathcal{Q}(\boldsymbol{z},\boldsymbol{y},\bar{\boldsymbol{y}},\bar{\boldsymbol{x}})-\Big[\mathcal{K}_{\boldsymbol{Y}\bar{\boldsymbol{Y}}}-\mathcal{K}_{\boldsymbol{X}\bar{\boldsymbol{Y}}}\Big]\,\mathcal{Q}(\boldsymbol{x},\boldsymbol{z},\bar{\boldsymbol{y}},\bar{\boldsymbol{x}})\,\mathcal{S}(\boldsymbol{z},\boldsymbol{y})\,.\end{aligned} \tag{6.8}$$

A few remarks are in order. First, eq. (6.8) emerges from our four-term colour combinations in a non-trivial way: within each channel, the no-scattering constants and the DA-only and CCA-only S-matrices cancel identically among the four terms, and 12 of the 16 elementary S-matrix

structures drop out; the surviving terms are those in which the soft gluon interacts (or is emitted close to the shockwave) on *both* sides of the cut, in one-to-one correspondence with the real terms of the B-JIMWLK equation for the quadrupole. Second — and this is the main point — eq. (6.8) is *identical* to the massless result, eq. (6.4) of [34]: neither the emission kernels nor the colour structures are affected by the quark mass, which enters eq. (6.7) only through the LO hard factor $\mathcal{H}^T_{\rm LO}$. Third, the second line of eq. (6.7) represents the action of the "production" version of the JIMWLK Hamiltonian [58, 59, 60] on the LO colour structure $\mathcal{W}_{\rm LO} = \mathcal{Q}(\boldsymbol{x},\boldsymbol{y},\bar{\boldsymbol{y}},\bar{\boldsymbol{x}}) - \mathcal{S}(\boldsymbol{x},\boldsymbol{y}) - \mathcal{S}(\bar{\boldsymbol{y}},\bar{\boldsymbol{x}}) + 1$ of the massive dijet cross-section (the pieces generated by the dipoles and the unit term are the subject of the virtual corrections, which are not considered here).

Finally, if the soft gluon is not measured, one integrates over its kinematics: the $\boldsymbol{k}_3$-integral sets $\bar{\boldsymbol{z}} = \boldsymbol{z}$, while $dk_3^+ = q^+ d\xi$ exhibits the logarithmic phase space $d\xi/\xi$. Recalling that $2\alpha_s C_F \simeq \alpha_s N_c \equiv \pi\bar{\alpha}_s$ at large N_c, the coefficient of the logarithm is precisely one step of the (real) B-JIMWLK evolution of the quadrupole [58, 61, 62],

$$\xi\,\frac{\partial\,\mathcal{Q}(\boldsymbol{x},\boldsymbol{y},\bar{\boldsymbol{y}},\bar{\boldsymbol{x}})}{\partial\xi}\bigg|_{\rm real} = \frac{\bar{\alpha}_s}{2\pi}\int_{\boldsymbol{z}}\Delta_{\rm real}\mathcal{Q}(\boldsymbol{x},\boldsymbol{y},\bar{\boldsymbol{y}},\bar{\boldsymbol{x}};\boldsymbol{z},\bar{\boldsymbol{z}}=\boldsymbol{z})\,, \tag{6.9}$$

in agreement with the real terms of the evolution equation for the quadrupole in refs. [58, 61]. We conclude that the slow-gluon logarithm generated by the massive trijet cross-section is subtracted and resummed by the B-JIMWLK (or, at large N_c, BK) evolution of the LO *massive* dijet cross-section, with an evolution kernel that is unmodified by the quark mass: all mass effects reside in the impact factor. This is as expected on physical grounds — the soft gluon has a short formation time $\tau_p \sim 2\xi q^+/\boldsymbol{p}^2$ and cannot resolve the quark mass, whose dead cone $Y \gtrsim (1-\zeta)/(\xi m_f)$ recedes to infinity as $\xi \to 0$ — but it is reassuring to see it emerge from the detailed structure of the massive helicity sums, where entire families of mass terms ($F^{(10)}$, $F^{(00)}$, all the single-flip interferences, and the mass corrections to the instantaneous terms) conspire to vanish in this limit.

7 Limits and consistency checks

7.1 The massless limit

For $m_f \to 0$ one has $\omega_f \to 0$, so all the hard-factor terms involving $F^{(10)}, F^{(01)}, F^{(00)}$ drop out of eqs. (4.4)–(4.6) (their coefficients are proportional to m_f and/or ω_f), and only the $F^{(11)}$ structure survives. Moreover, in this limit $A_u \to \epsilon_f^2 = \widetilde{Q}^2$ becomes independent of u, and the Feynman-parameter integral in eq. (3.8) can be done in closed form (cf. appendix A):

$$\mathcal{G}^{(11)}\Big|_{m_f=0} = \frac{\bar{Q}\,K_1\big(\bar{Q}\sqrt{R^2+aY^2}\big)}{\zeta(1-\zeta)\,Y^2\,\sqrt{R^2+aY^2}} = \frac{Q\,K_1(Q\,\mathcal{D})}{\mathcal{D}\,Y^2}\,, \tag{7.1}$$

with $\bar{Q}^2 \equiv \widetilde{Q}^2$ and where we used $\mathcal{D}^2 = \zeta(1-\zeta)\,(R^2+aY^2)$. Similarly $\mathcal{A}^{(11)} \to \bar{Q}K_1(\bar{Q}R)/[\zeta(1-\zeta)R] \times 1/Y^2$ and $\mathcal{G}_E \to QK_1(Q\mathcal{D})/\mathcal{D}$. Inserting eq. (7.1) into eqs. (3.11) and (4.2), and recalling that the massless limit of the first line of eq. (4.4) coincides with eq. (5.5) of [34], one verifies that our wave function and cross-section reproduce *exactly* — including all normalisation factors, cf. $\alpha_{em}\alpha_s = e^2g^2/(4(2\pi)^2)$ — the massless results in eqs. (4.25), (5.2), (5.15) of that reference, and thereby also the momentum-space results of refs. [38, 39] with which the latter agree.

7.2 Structure of the mass corrections

A few comments on the physics of the new terms are in order.

Helicity flips and azimuthal structure. The mass terms open helicity-flip channels at both vertices. In the direct terms these flips enter squared (e.g. $m_f^2\,F^{(01)}\bar{F}^{(01)}$) or as double-flip interferences (e.g. $m_f\xi\omega_f\,F^{(11)}\bar{F}^{(00)}$); both effects are of order m_f^2 and modify the angular correlations

between the produced partons. In particular, the terms proportional to $\epsilon^{ij}\epsilon^{mn}$ in eq. (4.4) — which generate $\cos 2\phi$-type azimuthal modulations — keep their massless coefficients, while the new scalar terms dilute them, in close analogy with the known mass effects in dijet azimuthal asymmetries [54].

The scales of the problem. The sizes of the two effective dipoles in the wave function are controlled by different scales: $R \lesssim 1/\epsilon_f$ for the $q\bar{q}$ system at the photon vertex and, for emissions after the collision, $Y \lesssim 1/\omega_f$ at the gluon vertex, cf. eq. (3.15). Since $\omega_f = \xi m_f/(1-\zeta)$ vanishes with ξ, the quark mass does not prevent a soft gluon from propagating far from its emitter: the dead-cone suppression sets in only for $Y \gtrsim (1-\zeta)/(\xi m_f)$. Prior to the collision, both scales combine into the Feynman-parameter structure of eq. (3.8), with the overall size of the three-parton system limited by $E_f\,\mathcal{D} \lesssim 1$, where the mass-shifted virtuality $E_f^2 = Q^2 + m_f^2/\zeta + m_f^2/\eta$ also cuts off the aligned-jet configurations $\zeta \to 0$ or $\eta \to 0$: unlike in the massless case, the transverse size of the pair remains bounded in these limits, which is the well-known taming of the aligned-jet configurations by the quark mass.

Soft gluon limit. For $\xi \to 0$ at fixed ζ the mass structures decouple and the cross-section reduces to the LO massive dijet cross-section times one step of the real B-JIMWLK evolution, with a kernel unmodified by the quark mass; this limit was analysed in detail in section 6.

Verification. All the algebraic ingredients of this paper — the spinor-matrix identities of section 2.2, the energy denominators, the identity (2.19), the Fourier transforms (3.8), (3.15) and (3.17), and every helicity sum in sections 4 and 5 and in appendix B — have been verified by independent symbolic computations, and the radial functions in addition by direct numerical integration of their defining two-dimensional Fourier transforms.

8 Conclusions

In this paper we have computed the three-parton ($q\bar{q}g$) component of the outgoing light-cone wave function of a transversely polarised virtual photon with *massive* quarks, together with the resulting leading-order cross-section for trijet production in DIS at small Bjorken x, in the CGC framework and in the large-N_c limit. The quark mass enriches the structure of the wave function in an essential way: both emission vertices acquire helicity-flip pieces, the single Bessel-function kernel of the massless case is replaced by four radial functions given by compact one-parameter integrals of Bessel functions, eq. (3.8), and the instantaneous pieces are governed by the mass-shifted virtuality $E_f^2 = Q^2 + m_f^2/\zeta + m_f^2/\eta$. The colour structure of the cross-section — dipoles and quadrupoles at large N_c — is untouched by the mass, so the new results are entirely contained in the hard factors, eqs. (4.4)–(4.6), (4.12)–(4.13) and (4.17)–(4.18), and in their momentum-space counterparts of section 5. In the massless limit we recover the known trijet results [34, 38, 39].

Our results admit several natural applications. First, they constitute the complete real-emission ingredient for the NLO corrections to *heavy-quark* dijet and dihadron production in DIS at small x: integrating out the kinematics of one of the three partons yields the real NLO corrections to the corresponding two-particle cross-sections, in direct analogy with the massless case [34, 32, 29]. In section 6 we took the first step in this direction: we showed that in the soft-gluon limit the trijet cross-section reduces to the LO massive dijet cross-section times the real part of the B-JIMWLK evolution, with the quark mass confined to the impact factor; the remaining ingredients are the finite (ξ-subtracted) real corrections and the massive virtual corrections, which can be adapted from refs. [44, 46]. Second, the trijet cross-section is interesting in its own right: three-particle azimuthal correlations with a tagged charm pair offer an independent handle on gluon saturation at the EIC, with the quark mass providing an additional lever arm in the comparison between light and heavy flavours. Finally, the momentum-space expressions

of section 5, which only involve low-dimensional convolutions with the dipole distribution, are directly amenable to numerical studies in the Gaussian approximation of appendix C; we plan to return to the associated phenomenology in future work.

Acknowledgements.

A Fourier transforms and the radial functions

In this appendix we derive the master formula (3.8). Combining the two denominators of eq. (3.7) with a double Schwinger parametrisation,

$$\frac{1}{\big(\widetilde{\boldsymbol{k}}^2+\epsilon_f^2\big)\big[\zeta(1-\zeta)^2\big(\widetilde{\boldsymbol{p}}^2+\omega_f^2\big)+\xi\eta\big(\widetilde{\boldsymbol{k}}^2+\epsilon_f^2\big)\big]}=\frac{1}{\xi\eta}\int_0^\infty ds\int_0^\infty dt\, e^{-(s+t)(\widetilde{\boldsymbol{k}}^2+\epsilon_f^2)-\frac{t}{a}(\widetilde{\boldsymbol{p}}^2+\omega_f^2)}\,, \tag{A.1}$$

with a from eq. (2.5), the transverse momentum integrations become Gaussian,

$$\int\frac{d^2\boldsymbol{k}}{2\pi}\,\{1,\,k^j\}\,e^{i\boldsymbol{k}\cdot\boldsymbol{R}-\sigma\boldsymbol{k}^2}=\Big\{\frac{1}{2\sigma},\,\frac{iR^j}{4\sigma^2}\Big\}\,e^{-R^2/4\sigma}\,, \tag{A.2}$$

and similarly for the $\widetilde{\boldsymbol{p}}$ integral (with $\boldsymbol{R}\to-\boldsymbol{Y}$ in the phase, whence the relative factors of $\pm i$ in eq. (3.7)). Changing variables from (s,t) to the overall scale $\sigma=s+t$ and the Feynman parameter $u=t/\sigma\in(0,1)$, the σ-integral is of the standard form

$$\int_0^\infty\frac{d\sigma}{\sigma^{\nu+1}}\,e^{-\sigma A-\frac{\Delta}{4\sigma}}=2\left(\frac{4A}{\Delta}\right)^{\nu/2}K_\nu\big(\sqrt{A\Delta}\big)\,, \tag{A.3}$$

and one arrives at eq. (3.8) with A_u and Δ_u as in eq. (3.9). The same manipulations with the single denominator of the instantaneous contribution, together with the identity (2.19) and $\mathcal{D}^2/[\zeta(1-\zeta)\xi\eta]=R^2/(\xi\eta)+Y^2/[\zeta(1-\zeta)^2]$, directly give eq. (3.17).

In the massless limit, $A_u\to\bar{Q}^2$ is u-independent and the integrals (3.10) can be done in closed form; for instance, using $\frac{d}{du}\big[\sqrt{A/\Delta_u}\,K_1(\sqrt{A\Delta_u})\big]=\frac{aY^2}{2u^2}\frac{A}{\Delta_u}K_2(\sqrt{A\Delta_u})/\ldots$ one readily recovers

$$\mathcal{G}^{(11)}\Big|_{m_f=0}=\frac{\bar{Q}\,K_1\big(\bar{Q}\sqrt{R^2+aY^2}\big)}{\zeta(1-\zeta)\,Y^2\sqrt{R^2+aY^2}}\,, \tag{A.4}$$

which is the coordinate-space kernel of [34], there obtained from the two-denominator transform

$$\begin{aligned}\int\frac{d^2\boldsymbol{k}}{2\pi}\,\frac{d^2\boldsymbol{p}}{2\pi}\,\frac{k^i\,p^j\;e^{i\boldsymbol{k}\cdot\boldsymbol{X}+i\boldsymbol{p}\cdot\boldsymbol{Z}}}{\big(\boldsymbol{k}^2+\bar{Q}^2\big)\big[a(\boldsymbol{k}^2+\bar{Q}^2)+\boldsymbol{p}^2\big]}&\\ =-\frac{X^iZ^j}{Z^2\sqrt{X^2+aZ^2}}\,\bar{Q}\,K_1\big(\bar{Q}\sqrt{X^2+aZ^2}\big)\,.&\end{aligned} \tag{A.5}$$

Two more limits are useful. (i) For $Y\to0$ at fixed R (the configuration relevant for the subtraction terms), $Y^2\,\mathcal{G}^{(11)}$ and $Y^2\,\mathcal{G}^{(01)}$ tend to their after-shockwave counterparts $Y^2\mathcal{A}^{(11)}$, $Y^2\mathcal{A}^{(01)}$, while $\mathcal{G}^{(10)}-\mathcal{A}^{(10)}$ and $\mathcal{G}^{(00)}-\mathcal{A}^{(00)}$ remain finite: the leading short-distance singularities of the emission are universal, as they must be. (ii) For $\omega_fY\gg1$ all four functions are exponentially suppressed, which is the coordinate-space manifestation of the dead cone.

B Helicity sums in momentum space

We collect here the complete set of helicity and polarisation sums of products of the numerator structures defined in section 5. They are the building blocks of the hard factors for arbitrary colour structures (elastic or not), and were used to obtain eqs. (5.17), (5.19) and (5.21). We use the notation $N(a)\equiv N(\boldsymbol{k}_a,\boldsymbol{p}_a)$ for the antiquark-emission numerator (5.10) evaluated on a generic pair of transverse momenta $(\boldsymbol{k}_a,\boldsymbol{p}_a)$, and $I^{lr}_{\lambda_1\lambda_2}\equiv(\delta^{lr}+2i\epsilon^{lr}\lambda_1)\delta_{\lambda_1\lambda_2}$, $\widetilde{I}^{lr}_{\lambda_1\lambda_2}\equiv(\delta^{lr}-2i\epsilon^{lr}\lambda_1)\delta_{\lambda_1\lambda_2}$ for the two instantaneous structures. All sums run over $\lambda_1,\lambda_2=\pm1/2$ and over the transverse indices l,r. With $\boldsymbol{k}_a\times\boldsymbol{k}_b\equiv\epsilon^{ij}k_a^ik_b^j$:

Antiquark emission.

$$\begin{aligned}\mathcal{K}(a;b) \equiv \sum N(a)N^*(b) &= 8\big[\zeta^2+(1-\zeta)^2\big]\big[\eta^2+(1-\zeta)^2\big](\boldsymbol{k}_a\cdot\boldsymbol{k}_b)(\boldsymbol{p}_a\cdot\boldsymbol{p}_b)\\ &\quad -8\,\xi(2\zeta-1)\big(2(1-\zeta)-\xi\big)(\boldsymbol{k}_a\times\boldsymbol{k}_b)(\boldsymbol{p}_a\times\boldsymbol{p}_b)\\ &\quad +8m_f^2\big[\eta^2+(1-\zeta)^2\big](\boldsymbol{p}_a\cdot\boldsymbol{p}_b)+8\xi^2\omega_f^2\big[\zeta^2+(1-\zeta)^2\big](\boldsymbol{k}_a\cdot\boldsymbol{k}_b)+8m_f^2\xi^2\omega_f^2\\ &\quad -8\,\zeta\eta\,m_f\xi\omega_f\,(\boldsymbol{k}_a-\boldsymbol{k}_b)\cdot(\boldsymbol{p}_a-\boldsymbol{p}_b)\,,\\ \mathcal{J}(a) \equiv \sum N(a)\,I^* &= 8\,\zeta\eta\,(\boldsymbol{k}_a\cdot\boldsymbol{p}_a)-8\,m_f\,\xi\,\omega_f\,, \qquad \sum I\,I^*=8\,.\end{aligned} \tag{B.1}$$

Quark emission.

$$\begin{aligned}\widetilde{\mathcal{K}}(a;b) &= 8\big[\zeta^2+(1-\eta)^2\big]\big[\eta^2+(1-\eta)^2\big](\boldsymbol{k}_a\cdot\boldsymbol{k}_b)(\boldsymbol{p}_a\cdot\boldsymbol{p}_b)\\ &\quad -8\,\xi(2\eta-1)\big(2(1-\eta)-\xi\big)(\boldsymbol{k}_a\times\boldsymbol{k}_b)(\boldsymbol{p}_a\times\boldsymbol{p}_b)\\ &\quad +8m_f^2\big[\zeta^2+(1-\eta)^2\big](\boldsymbol{p}_a\cdot\boldsymbol{p}_b)+8\xi^2\widetilde{\omega}_f^2\big[\eta^2+(1-\eta)^2\big](\boldsymbol{k}_a\cdot\boldsymbol{k}_b)+8m_f^2\xi^2\widetilde{\omega}_f^2\\ &\quad +8\,\zeta\eta\,m_f\xi\widetilde{\omega}_f\,(\boldsymbol{k}_a-\boldsymbol{k}_b)\cdot(\boldsymbol{p}_a-\boldsymbol{p}_b)\,,\\ \widetilde{\mathcal{J}}(a) \equiv \sum \widetilde{N}(a)\widetilde{I}^* &= -8\,\zeta\eta\,(\boldsymbol{k}_a\cdot\boldsymbol{p}_a)-8\,m_f\,\xi\,\widetilde{\omega}_f\,, \qquad \sum \widetilde{I}\,\widetilde{I}^*=8\,.\end{aligned} \tag{B.2}$$

Interference.

$$\begin{aligned}\mathcal{K}^{\times}(a;b) \equiv \sum \widetilde{N}(a)N^*(b) &= 8\big[1-\xi-2\zeta\eta\big]\big[\xi(1-\xi)+2\zeta\eta\big](\boldsymbol{k}_a\cdot\boldsymbol{k}_b)(\boldsymbol{p}_a\cdot\boldsymbol{p}_b)\\ &\quad -8\,\xi\big[1-2\zeta-\xi\big]^2(\boldsymbol{k}_a\times\boldsymbol{k}_b)(\boldsymbol{p}_a\times\boldsymbol{p}_b)\\ &\quad +8m_f^2(\xi+2\zeta\eta)(\boldsymbol{p}_a\cdot\boldsymbol{p}_b)-8(\xi+2\zeta\eta)\,\xi^2\widetilde{\omega}_f\omega_f\,(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)\\ &\quad -8(\zeta+\xi)^2m_f\xi\omega_f\,(\boldsymbol{k}_a\cdot\boldsymbol{p}_a)\\ &\quad +8(1-\zeta)^2m_f\xi\widetilde{\omega}_f\,(\boldsymbol{k}_b\cdot\boldsymbol{p}_b)-8\,\eta^2m_f\xi\widetilde{\omega}_f\,(\boldsymbol{k}_a\cdot\boldsymbol{p}_b)+8\,\zeta^2m_f\xi\omega_f\,(\boldsymbol{p}_a\cdot\boldsymbol{k}_b)\,,\\ \mathcal{J}^{\times}(a) \equiv \sum \widetilde{N}(a)\,I^* &= 8(\zeta+\xi)^2(\boldsymbol{k}_a\cdot\boldsymbol{p}_a)\,,\\ \bar{\mathcal{J}}^{\times}(b) \equiv \sum \widetilde{I}\,N^*(b) &= -8(1-\zeta)^2(\boldsymbol{k}_b\cdot\boldsymbol{p}_b)\,, \qquad \sum \widetilde{I}\,I^*=0\,.\end{aligned} \tag{B.3}$$

Setting $m_f=0$, $\mathcal{K}$, $\widetilde{\mathcal{K}}$ and $\mathcal{K}^{\times}$ reduce to the massless squared vertices of ref. [34] (their eqs. (5.5) and (5.17)).

C Gaussian approximation for the colour correlators

For numerical applications, the quadrupoles appearing in eqs. (4.9)–(4.10) and (4.19) may be evaluated in the Gaussian approximation for the CGC target averages [62, 63, 52], in which all correlators of Wilson lines are expressed in terms of the dipole $\mathcal{S}(\boldsymbol{x},\boldsymbol{y})=\mathrm{e}^{-\Gamma(\boldsymbol{x},\boldsymbol{y})}$. At large N_c the quadrupole reads

$$\mathcal{Q}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3,\boldsymbol{x}_4)=\mathcal{S}(\boldsymbol{x}_1,\boldsymbol{x}_2)\,\mathcal{S}(\boldsymbol{x}_3,\boldsymbol{x}_4)-\frac{F(\boldsymbol{x}_1,\boldsymbol{x}_2;\boldsymbol{x}_3,\boldsymbol{x}_4)}{F(\boldsymbol{x}_1,\boldsymbol{x}_3;\boldsymbol{x}_2,\boldsymbol{x}_4)}\Big[\mathcal{S}(\boldsymbol{x}_1,\boldsymbol{x}_2)\,\mathcal{S}(\boldsymbol{x}_3,\boldsymbol{x}_4)-\mathcal{S}(\boldsymbol{x}_1,\boldsymbol{x}_4)\,\mathcal{S}(\boldsymbol{x}_3,\boldsymbol{x}_2)\Big]\,, \tag{C.1}$$

with

$$F(\boldsymbol{x}_1,\boldsymbol{x}_2;\boldsymbol{x}_3,\boldsymbol{x}_4)\equiv\Gamma(\boldsymbol{x}_1,\boldsymbol{x}_3)+\Gamma(\boldsymbol{x}_2,\boldsymbol{x}_4)-\Gamma(\boldsymbol{x}_1,\boldsymbol{x}_4)-\Gamma(\boldsymbol{x}_2,\boldsymbol{x}_3)\,. \tag{C.2}$$

One easily checks the coincidence limits $\mathcal{Q}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_2,\boldsymbol{x}_4)=\mathcal{S}(\boldsymbol{x}_1,\boldsymbol{x}_4)$ and $\mathcal{Q}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3,\boldsymbol{x}_3)=\mathcal{S}(\boldsymbol{x}_1,\boldsymbol{x}_2)$. Inserting eq. (C.1) into the large-N_c colour structures of section 4 gives, for the antiquark-emission direct terms,

$$\begin{aligned}
\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) &= \Big[\mathcal{S}_{xz}\mathcal{S}_{\bar z\bar x}-\frac{F(\boldsymbol{x},\boldsymbol{z};\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})}{F(\boldsymbol{x},\bar{\boldsymbol{z}};\boldsymbol{z},\bar{\boldsymbol{x}})}\big[\mathcal{S}_{xz}\mathcal{S}_{\bar z\bar x}-\mathcal{S}_{x\bar x}\mathcal{S}_{\bar z z}\big]\Big]\\
&\quad\times\Big[\mathcal{S}_{zy}\mathcal{S}_{\bar y\bar z}-\frac{F(\boldsymbol{z},\boldsymbol{y};\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})}{F(\boldsymbol{z},\bar{\boldsymbol{y}};\boldsymbol{y},\bar{\boldsymbol{z}})}\big[\mathcal{S}_{zy}\mathcal{S}_{\bar y\bar z}-\mathcal{S}_{z\bar z}\mathcal{S}_{\bar y y}\big]\Big]\\
&\quad-\mathcal{S}_{xz}\mathcal{S}_{zy}-\mathcal{S}_{\bar z\bar x}\mathcal{S}_{\bar y\bar z}+1\,,\\
\mathcal{W}(\boldsymbol{x},\boldsymbol{y}',\boldsymbol{y}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) &= \Big[\mathcal{S}_{xy'}\mathcal{S}_{\bar z\bar x}-\frac{F(\boldsymbol{x},\boldsymbol{y}';\bar{\boldsymbol{z}},\bar{\boldsymbol{x}})}{F(\boldsymbol{x},\bar{\boldsymbol{z}};\boldsymbol{y}',\bar{\boldsymbol{x}})}\big[\mathcal{S}_{xy'}\mathcal{S}_{\bar z\bar x}-\mathcal{S}_{x\bar x}\mathcal{S}_{\bar z y'}\big]\Big]\mathcal{S}_{\bar y\bar z}\\
&\quad-\mathcal{S}_{xy'}-\mathcal{S}_{\bar z\bar x}\mathcal{S}_{\bar y\bar z}+1\,,\\
\mathcal{W}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z};\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}',\bar{\boldsymbol{y}}') &= \Big[\mathcal{S}_{xz}\mathcal{S}_{\bar y'\bar x}-\frac{F(\boldsymbol{x},\boldsymbol{z};\bar{\boldsymbol{y}}',\bar{\boldsymbol{x}})}{F(\boldsymbol{x},\bar{\boldsymbol{y}}';\boldsymbol{z},\bar{\boldsymbol{x}})}\big[\mathcal{S}_{xz}\mathcal{S}_{\bar y'\bar x}-\mathcal{S}_{x\bar x}\mathcal{S}_{\bar y' z}\big]\Big]\mathcal{S}_{zy}\\
&\quad-\mathcal{S}_{xz}\mathcal{S}_{zy}-\mathcal{S}_{\bar y'\bar x}+1\,,\\
\mathcal{W}(\boldsymbol{x},\boldsymbol{y}',\boldsymbol{y}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}',\bar{\boldsymbol{y}}') &= \mathcal{S}_{xy'}\mathcal{S}_{\bar y'\bar x}-\mathcal{S}_{xy'}-\mathcal{S}_{\bar y'\bar x}+1\\
&\quad-\frac{F(\boldsymbol{x},\boldsymbol{y}';\bar{\boldsymbol{y}}',\bar{\boldsymbol{x}})}{F(\boldsymbol{x},\bar{\boldsymbol{y}}';\boldsymbol{y}',\bar{\boldsymbol{x}})}\big[\mathcal{S}_{xy'}\mathcal{S}_{\bar y'\bar x}-\mathcal{S}_{x\bar x}\mathcal{S}_{\bar y' y'}\big]\,,
\end{aligned}\tag{C.3}$$

where $\mathcal{S}_{ab}\equiv\mathcal{S}(\boldsymbol{a},\boldsymbol{b})$. The quark-emission direct terms are obtained by the relabelling $(\boldsymbol{x},\boldsymbol{y},\boldsymbol{y}')\to(\boldsymbol{y},\boldsymbol{x},\boldsymbol{x}')$ (and likewise for the barred coordinates), e.g.

$$\mathcal{W}(\boldsymbol{y},\boldsymbol{x}',\boldsymbol{x}';\bar{\boldsymbol{y}},\bar{\boldsymbol{x}},\bar{\boldsymbol{z}})=\Big[\mathcal{S}_{yx'}\mathcal{S}_{\bar z\bar y}-\frac{F(\boldsymbol{y},\boldsymbol{x}';\bar{\boldsymbol{z}},\bar{\boldsymbol{y}})}{F(\boldsymbol{y},\bar{\boldsymbol{z}};\boldsymbol{x}',\bar{\boldsymbol{y}})}\big[\mathcal{S}_{yx'}\mathcal{S}_{\bar z\bar y}-\mathcal{S}_{y\bar y}\mathcal{S}_{\bar z x'}\big]\Big]\mathcal{S}_{\bar x\bar z}-\mathcal{S}_{yx'}-\mathcal{S}_{\bar z\bar y}\mathcal{S}_{\bar x\bar z}+1\,,\tag{C.4}$$

while for the interference one needs in addition

$$\begin{aligned}
\mathcal{W}(\boldsymbol{x}',\boldsymbol{y},\boldsymbol{x}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}},\bar{\boldsymbol{z}}) &= \mathcal{S}_{\bar z\bar x}\Big[\mathcal{S}_{x'y}\mathcal{S}_{\bar y\bar z}-\frac{F(\boldsymbol{x}',\boldsymbol{y};\bar{\boldsymbol{y}},\bar{\boldsymbol{z}})}{F(\boldsymbol{x}',\bar{\boldsymbol{y}};\boldsymbol{y},\bar{\boldsymbol{z}})}\big[\mathcal{S}_{x'y}\mathcal{S}_{\bar y\bar z}-\mathcal{S}_{x'\bar z}\mathcal{S}_{\bar y y}\big]\Big]-\mathcal{S}_{x'y}-\mathcal{S}_{\bar z\bar x}\mathcal{S}_{\bar y\bar z}+1\,,\\
\mathcal{W}(\boldsymbol{x}',\boldsymbol{y},\boldsymbol{x}';\bar{\boldsymbol{x}},\bar{\boldsymbol{y}}',\bar{\boldsymbol{y}}') &= \mathcal{S}_{\bar y'\bar x}\,\mathcal{S}_{x'y}-\mathcal{S}_{x'y}-\mathcal{S}_{\bar y'\bar x}+1\,,
\end{aligned}\tag{C.5}$$

the last structure being exact (it contains no quadrupole). Keeping only the first, factorised term in each square bracket yields the "elastic" approximation used in section 5.

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